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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 7

FURTHER CURVE SKETCHING AND INEQUALITIES

SUGGESTED TIME: 12 PERIODS

IN THIS CHAPTER

This chapter completes the study of functions begun in Lower Sixth: domains, limits, asymptotes, stationary points, concavity and symmetry are combined into a systematic method for sketching curves. It also develops algebraic and graphical techniques for solving absolute-value inequalities, a favourite GCE Advanced Level topic.

- By the end of this chapter I can determine the domain of definition of any algebraic, rational, irrational or composite function.
- By the end of this chapter I can compute limits at the bounds of the domain, including infinite limits and limits at infinity.
- By the end of this chapter I can find the range of values of a parameter for which a function (or equation) exists or has real solutions.
- By the end of this chapter I can identify vertical, horizontal and oblique asymptotes and the intercepts of a curve.
- By the end of this chapter I can locate stationary points and determine their nature using the first or second derivative.
- By the end of this chapter I can construct a complete table of variation from the sign of the derivative.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Domain of Definition and Limits at the Bounds
2. Asymptotes, Intercepts and Infinite Branches
3. Stationary Points and Variation of a Function
4. Concavity, Points of Inflexion and Symmetry
5. Algebraic Solutions of Absolute-Value Inequalities
6. Graphical Solutions of Inequalities and Complete Curve Sketching
7. Integration activity
8. Methods and worked examples
9. Exercises
10. Solutions to the exercises
11. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can determine the domain of definition of any algebraic, rational, irrational or composite function.
- By the end of this chapter I can compute limits at the bounds of the domain, including infinite limits and limits at infinity.
- By the end of this chapter I can find the range of values of a parameter for which a function (or equation) exists or has real solutions.
- By the end of this chapter I can identify vertical, horizontal and oblique asymptotes and the intercepts of a curve.
- By the end of this chapter I can locate stationary points and determine their nature using the first or second derivative.
- By the end of this chapter I can construct a complete table of variation from the sign of the derivative.
- By the end of this chapter I can study the position of a curve relative to a line and describe infinite branches.
- By the end of this chapter I can determine concavity and points of inflexion using the second derivative.
- By the end of this chapter I can solve absolute-value inequalities both algebraically and graphically.

- By the end of this chapter I can detect centres and axes of symmetry and produce a full, justified sketch of a curve.

PREREQUISITES

- Differentiation of polynomial, rational and composite functions (Lower Sixth).
- Basic limits and continuity of a function at a point (Lower Sixth).
- Solving linear and quadratic equations and inequalities.
- The modulus (absolute value) function and its graph.
- Equation of a straight line and coordinate geometry basics.

1 DOMAIN OF DEFINITION AND LIMITS AT THE BOUNDS

Imagine a mobile money agent in Bamenda. Her daily profit, modelled by a rational function of the number of transactions she processes, rises steeply at first, then levels off, and never exceeds a certain ceiling no matter how hard she works. Why does the profit curve flatten towards a horizontal line, and where exactly does it bend? How can she know for which numbers of transactions her profit stays above a fixed threshold — that is, how can she solve an inequality involving absolute values of her daily balance? Curve sketching and inequalities give the complete answer: they let us predict the whole behaviour of a function before plotting a single point. This chapter turns you into a mathematical detective, able to draw any reasonable curve and to read off solutions of inequalities from it.

Every investigation of a curve begins with two fundamental questions: Where is the function defined? and What happens at the edges of that territory? That is exactly the business of this section.

1.1 The Domain of Definition of a Function

Intuition first. A function is a machine: you feed it a number x, and it produces a number $f(x)$. But no machine accepts every input. You cannot divide by zero, you cannot take the square root of a negative quantity (in $\mathbb{R}$), and you cannot take the logarithm of a negative number or of zero. The **domain of definition** is simply the list of all inputs the machine will accept.

DEFINITION **Domain of definition**

Let f be a real-valued function of a real variable. The **domain of definition** of f, written D_f, is the set of all real numbers x for which $f(x)$ exists and is a real number:

$$D_f = \{\, x \in \mathbb{R} \;:\; f(x) \text{ exists in } \mathbb{R} \,\}.$$

PROPERTY **Standard domain conditions**

When searching for D_f, apply the following rules.

1. A denominator must be non-zero: in $\dfrac{g(x)}{h(x)}$, we require $h(x) \neq 0$.
2. The radicand of an even root must be non-negative: in $\sqrt{u(x)}$, we require $u(x) \geq 0$.
3. (Useful extra for the GCE) The argument of a logarithm must be strictly positive: in $\ln\big(u(x)\big)$, we require $u(x) > 0$.

WARNING **Square root in a denominator**

A very common mistake: for $\frac{1}{\sqrt{u(x)}}$, students write only $u(x) \geq 0$. This is wrong! The radicand must be non-negative and the denominator must be non-zero, so the correct condition is $u(x) > 0$ strictly. For example, $f(x) = \frac{1}{\sqrt{x-2}}$ has domain $]2, +\infty[$, not $[2, +\infty[$.

METHOD **Finding the domain of definition**

1. List the constraints: identify every denominator, every even root, every logarithm in the expression of f.
2. Solve each constraint separately (equations or inequalities in x).
3. Intersect the solution sets: x must satisfy all constraints simultaneously.
4. Write the answer as a union of intervals, using brackets [ ] for included bounds and] [for excluded bounds.

Domains of combined functions. If f and g have domains D_f and D_g, then:

- the **sum** $f + g$ and the **product** fg are defined on $D_f \cap D_g$;
- the **quotient** $\frac{f}{g}$ is defined on $\{ x \in D_f \cap D_g : g(x) \neq 0 \}$;
- the **composite** $g \circ f$ is defined on $\{ x \in D_f : f(x) \in D_g \}$: x must be acceptable to f, and the output $f(x)$ must be acceptable to g.

WORKED EXAMPLE 1 **Domain of a mixed function**

Determine the domain of definition of $f(x) = \frac{\sqrt{x-1}}{x-3}$.

SOLUTION

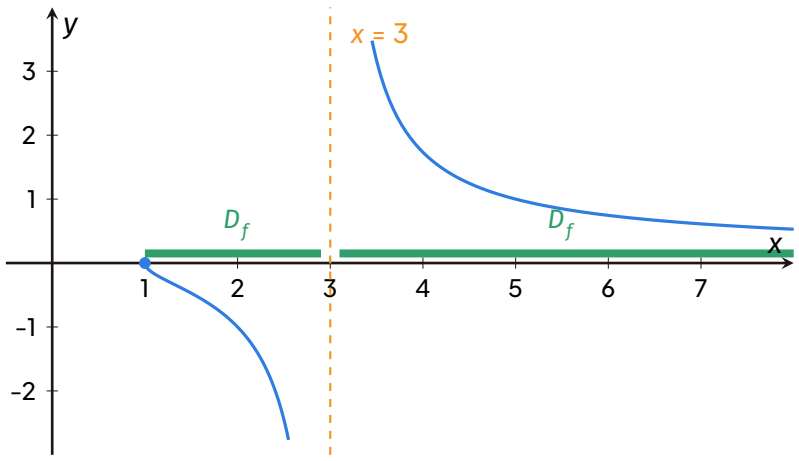
Solution. We list the constraints.

- Radicand: $x - 1 \geq 0 \iff x \geq 1$.
- Denominator: $x - 3 \neq 0 \iff x \neq 3$.

Intersecting: $x \geq 1$ and $x \neq 3$, hence

$$D_f = [1, 3[\cup]3, +\infty[.$$

Note that $x = 1$ is included (the numerator may be zero), while $x = 3$ is excluded.



Graph of $f(x) = \dfrac{\sqrt{x-1}}{x-3}$. The domain $[1,3[\cup]3,+\infty[$ is highlighted in green on the x-axis; the dashed line $x = 3$ is excluded from the domain.

1.2 Limits at the Bounds of the Domain

Once D_f is known, the next detective question is: how does f behave as x approaches a bound of D_f? The bounds are of two kinds: **finite bounds** (like 1 and 3 above) and the **infinite bounds** $-\infty$ and $+\infty$.

DEFINITION **Finite limit at a point; one-sided limits**

We say that $f(x)$ tends to the real number ℓ as x tends to a, and we write $\lim_{x \to a} f(x) = \ell$, if $f(x)$ becomes and stays arbitrarily close to ℓ whenever x is close enough to a (with $x \neq a$).

- The **right-hand limit** $\lim_{x \to a^+} f(x)$ uses only values $x > a$.
- The **left-hand limit** $\lim_{x \to a^-} f(x)$ uses only values $x < a$.

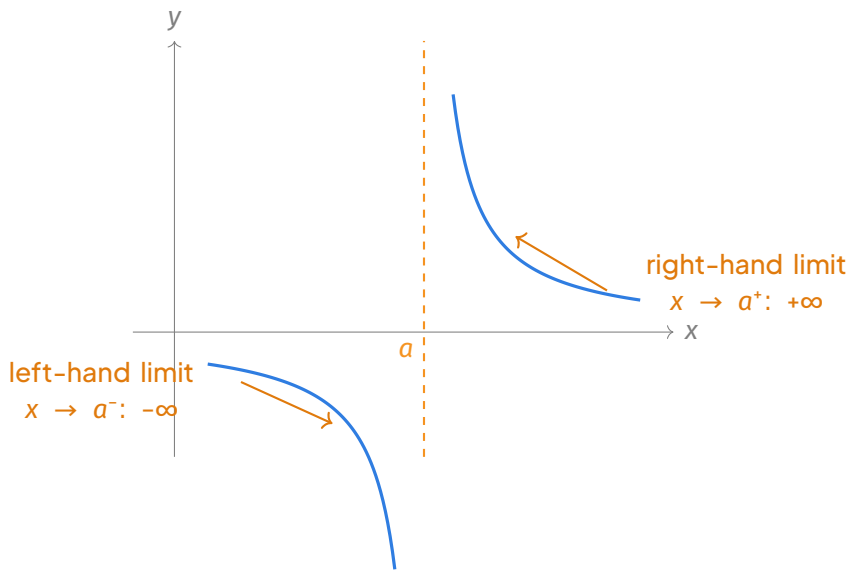
The limit $\lim_{x \to a} f(x)$ exists if and only if both one-sided limits exist and are equal.

WARNING **Compare both sides before concluding**

A classic trap: students compute one side, find $+\infty$, and stop. You must always compare the left-hand and right-hand limits. If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$, then $\lim_{x \to a} f(x)$ does not exist. Example: for $f(x) = \dfrac{1}{x-3}$, the left-hand limit is $-\infty$ and the right-hand limit is $+\infty$; the two-sided limit does not exist, even though each side has a perfectly good infinite limit.

PROPERTY **Infinite limits and vertical asymptotes**

If $\lim_{x \to a^+} f(x) = \pm\infty$ or $\lim_{x \to a^-} f(x) = \pm\infty$, then the line $x = a$ is a **vertical asymptote** of the curve of f. This typically occurs when the denominator vanishes at a while the numerator does not.



Left-hand and right-hand limits at a bound a: the arrows show the direction of approach along each branch of the curve.

WORKED EXAMPLE 2 **One-sided limits at a finite bound**

For $f(x) = \dfrac{\sqrt{x-1}}{x-3}$, compute the limits at the bounds $x = 1$ and $x = 3$.

SOLUTION

Solution. At $x = 1$: only the right-hand limit makes sense (the domain starts at 1). As $x \to 1^+$, $\sqrt{x-1} \to 0$ and $x - 3 \to -2 \neq 0$, so

$$\lim_{x \to 1^+} f(x) = \frac{0}{-2} = 0.$$

The curve starts at the point $(1, 0)$. At $x = 3$: the numerator tends to $\sqrt{2} > 0$. For the denominator, use a sign table of $x - 3$: it is negative for $x < 3$ and positive for $x > 3$. Hence

$$\lim_{x \to 3^-} f(x) = \frac{\sqrt{2}}{0^-} = -\infty, \qquad \lim_{x \to 3^+} f(x) = \frac{\sqrt{2}}{0^+} = +\infty.$$

The line $x = 3$ is a vertical asymptote, and the two-sided limit at 3 does not exist.

1.3 Limits at Infinity: Polynomials, Rational Functions, Radicals

Polynomials. At $\pm\infty$, a polynomial behaves like its highest-degree term, because that term eventually dwarfs all the others:

$$\lim_{x\to+\infty}\left(2x^3-5x^2+7\right)=\lim_{x\to+\infty}2x^3=+\infty.$$

PROPERTY **Rational functions at infinity**

Let $f(x)=\dfrac{P(x)}{Q(x)}$ where P and Q are polynomials. Then at $+\infty$ or $-\infty$, the limit of f equals the limit of the ratio of the dominant (highest-degree) terms:

$$\lim_{x\to\pm\infty}\frac{a_nx^n+\cdots}{b_mx^m+\cdots}=\lim_{x\to\pm\infty}\frac{a_nx^n}{b_mx^m}.$$

Consequently the limit is: 0 if $\deg P<\deg Q$; the ratio $\dfrac{a_n}{b_m}$ of leading coefficients if $\deg P=\deg Q$; and $\pm\infty$ (with sign to be determined) if $\deg P>\deg Q$. (Proof by factorising the dominant term — see the method below; it is instructive and frequently demanded in examinations.)

Why factorisation works. Take $f(x)=\dfrac{3x^2-x+1}{2x^2+5}$. For $x\neq0$, factor the dominant term top and bottom:

$$f(x)=\frac{x^2\left(3-\frac{1}{x}+\frac{1}{x^2}\right)}{x^2\left(2+\frac{5}{x^2}\right)}=\frac{3-\frac{1}{x}+\frac{1}{x^2}}{2+\frac{5}{x^2}}\xrightarrow[x\to\pm\infty]{}\frac{3}{2},$$

since $\dfrac{1}{x}\to0$ and $\dfrac{1}{x^2}\to0$. This is exactly the mobile money agent's "ceiling": a finite limit ℓ at infinity means the curve flattens towards the horizontal line $y=\ell$.

METHOD **Removing an indeterminate form**

The **indeterminate forms** are $\dfrac{0}{0}$, $\dfrac{\infty}{\infty}$, $0\times\infty$, $\infty-\infty$ (and powers such as 1^∞). They mean "the rules of limits do not apply directly — transform the expression".

1. Form $\dfrac{\infty}{\infty}$ at infinity (polynomials, rational functions): factorise the dominant term in the numerator and in the denominator, simplify, then pass to the limit.
2. Form $\dfrac{0}{0}$ at a finite point a: factorise $(x-a)$ in numerator and denominator and cancel.
3. Form $\infty-\infty$ with radicals: multiply by the conjugate: use $\sqrt{A}-\sqrt{B}=\dfrac{A-B}{\sqrt{A}+\sqrt{B}}$.

WORKED EXAMPLE 3 **Conjugate multiplication**

Compute $\lim_{x\to+\infty}\left(\sqrt{x^2+3x}-x\right)$.

SOLUTION

Solution. This has the indeterminate form $\infty-\infty$. Multiply by the conjugate:

$$\sqrt{x^2+3x}-x=\frac{\left(\sqrt{x^2+3x}-x\right)\left(\sqrt{x^2+3x}+x\right)}{\sqrt{x^2+3x}+x}=\frac{x^2+3x-x^2}{\sqrt{x^2+3x}+x}=\frac{3x}{\sqrt{x^2+3x}+x}.$$

For $x>0$, $\sqrt{x^2+3x}=x\sqrt{1+\frac{3}{x}}$, so

$$\frac{3x}{x\sqrt{1+\frac{3}{x}}+x}=\frac{3}{\sqrt{1+\frac{3}{x}}+1}\xrightarrow[x\to+\infty]{}\frac{3}{2}.$$

Hence $\lim_{x\to+\infty}\left(\sqrt{x^2+3x}-x\right)=\frac{3}{2}$.

WARNING **Radicals and the sign of x**

When $x\to-\infty$, $\sqrt{x^2}=|x|=-x$, not x. Forgetting this changes signs and ruins the limit. Always write $\sqrt{x^2}=|x|$ first, then discuss the sign of x.

1.4 Range of Values of a Parameter for Existence of a Function

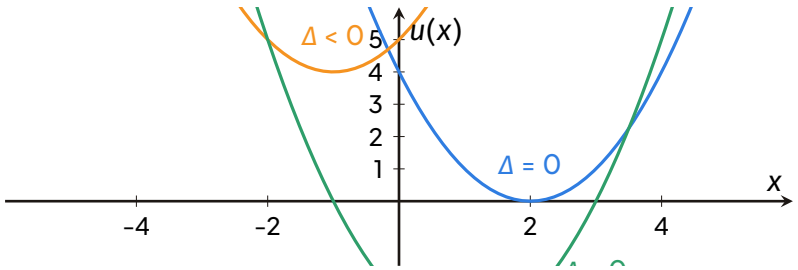
Many GCE questions ask: for which values of the parameter m is the function defined for all real x (or: on a given interval)? The key tool is the sign of a quadratic, controlled by its discriminant.

PROPERTY **Sign of a quadratic and the discriminant**

Let $u(x)=ax^2+bx+c$ with $a\neq0$ and discriminant $\Delta=b^2-4ac$.

- If $\Delta<0$, then $u(x)$ keeps the constant sign of a on $\mathbb{R}$.
- If $\Delta=0$, then $u(x)$ has the sign of a and vanishes once (at $x=-\frac{b}{2a}$).
- If $\Delta>0$, then $u(x)$ has the sign of a outside the roots and the opposite sign between them.

In particular, $u(x)\geq0$ for all real x if and only if $a>0$ and $\Delta\leq0$.



Sign of a quadratic according to its discriminant Δ.

WORKED EXAMPLE 4 **Parameter condition for existence on $\mathbb{R}$**

Find all values of the real parameter m for which $f(x) = \sqrt{x^2 + mx + 4}$ is defined for every real number x.

SOLUTION

Solution. We need the radicand $u(x) = x^2 + mx + 4$ to satisfy $u(x) \geq 0$ for all $x \in \mathbb{R}$. Here $a = 1 > 0$, so by the property above it is necessary and sufficient that $\Delta \leq 0$:

$$\Delta = m^2 - 16 \leq 0 \iff m^2 \leq 16 \iff -4 \leq m \leq 4.$$

Hence f is defined on the whole of $\mathbb{R}$ exactly when $m \in [-4, 4]$. For $|m| > 4$, the radicand becomes negative between the two roots of u, and the domain of f is only a union of two unbounded intervals: the function fails to exist on a whole central band.
Sign table for $u(x) = x^2 + mx + 4$ when $m = 5$ ($\Delta > 0$):

x	$-\infty$		x_1		x_2		$+\infty$		
$u(x)$	+	↘	0	↘	−	↗	0	↗	+

where x_1, x_2 are the roots. The function is undefined on $]x_1, x_2[$.

EXERCISE 1

1. Determine the domain of definition of $g(x) = \dfrac{\sqrt{4 - x^2}}{x + 1}$ and of $h(x) = \ln\left(\dfrac{x - 1}{x + 2}\right)$.
2. Compute $\lim_{x \to 2^-} \dfrac{x + 1}{x^2 - 4}$ and $\lim_{x \to 2^+} \dfrac{x + 1}{x^2 - 4}$. Does the two-sided limit exist?
3. Compute $\lim_{x \to -\infty} \dfrac{2x^2 - 3x}{x^2 + 1}$ and $\lim_{x \to +\infty} \left(\sqrt{x^2 + x} - x\right)$.

4. Find the values of k for which $f(x) = \dfrac{1}{\sqrt{kx^2 - 4x + k}}$ is defined for every real x.

SOLUTION **Exercise 1**

1. For g: $4 - x^2 \geq 0 \iff -2 \leq x \leq 2$, and $x \neq -1$. Hence $D_g = [-2, -1[\cup]-1, 2]$. For h: we need $\dfrac{x-1}{x+2} > 0$; a sign table gives $D_h =]-\infty, -2[\cup]1, +\infty[$.
2. $x^2 - 4 = (x-2)(x+2)$. Near $x = 2$ the numerator tends to $3 > 0$. For $x \to 2^-$: $(x-2) < 0$, $(x+2) > 0$, so the denominator is 0^- and the limit is $-\infty$. For $x \to 2^+$: denominator 0^+, limit $+\infty$. The two one-sided limits differ, so the two-sided limit does not exist; $x = 2$ is a vertical asymptote.
3. Ratio of dominant terms: $\dfrac{2x^2}{x^2} = 2$, so the limit is 2 (note: the sign of x does not matter here since the degrees are even and equal). For the radical: conjugate multiplication gives $\dfrac{x}{\sqrt{x^2+x}+x} = \dfrac{1}{\sqrt{1+\frac{1}{x}}+1} \to \dfrac{1}{2}$.
4. We need $kx^2 - 4x + k > 0$ for all x (strict, because of the denominator). Conditions: $k > 0$ and $\Delta = 16 - 4k^2 < 0$, i.e. $k^2 > 4$. Combined with $k > 0$: $k > 2$.

KEY POINTS **Key points of this section**

- D_f is found by listing constraints (denominator $\neq 0$; even radicand ≥ 0; logarithm argument > 0), solving each, and intersecting; write the result as a union of intervals.
- A two-sided limit exists only when the left-hand and right-hand limits exist and are equal.
- An infinite one-sided limit at $x = a$ signals a vertical asymptote $x = a$.
- At infinity, keep only the dominant terms; remove indeterminate forms by factorising the dominant term or multiplying by the conjugate.
- A radicand $ax^2 + bx + c$ stays non-negative on $\mathbb{R}$ exactly when $a > 0$ and $\Delta \leq 0$ — the standard tool for parameter questions on existence.

2 ASYMPTOTES, INTERCEPTS AND INFINITE BRANCHES

In the previous section we learnt how to determine the **domain of definition** of a function and how to compute its **limits at the bounds** of that domain. Those limits are not merely numbers:

they carry a precise geometric meaning. When a limit at a finite point is infinite, the curve climbs or dives without bound near a vertical line; when a limit at infinity is finite, the curve settles down along a horizontal line. This section turns those analytic facts into geometry: we define **asymptotes** and **intercepts**, we classify all possible **infinite branches**, and we learn how to determine the **position of a curve with respect to a line**. These are the tools that will make our curve sketching, at the end of the chapter, fast and reliable.

2.1 Intercepts of a Curve

DEFINITION **Intercepts**

Let f be a function and $\mathcal{C}_f$ its curve in an orthonormal frame $(O;\vec{\imath},\vec{\jmath})$.

- The **y-intercept** is the point where $\mathcal{C}_f$ meets the y-axis. If 0 belongs to the domain of f, it is the point $(0,\ f(0))$.
- The **x-intercepts** are the points where $\mathcal{C}_f$ meets the x-axis. Their abscissae are the solutions of the equation $f(x) = 0$ that belong to the domain of f.

The intercepts are the first points we plot: they anchor the curve to the axes. Note that a curve has at most one y-intercept (a function assigns a single image to 0), but it may have several x-intercepts, or none at all.

EXEMPLE **Finding intercepts**

Let $f(x) = \dfrac{x^2 - 4}{x + 1}$, defined on $\mathbb{R} \setminus \{-1\}$.

- y-intercept: $f(0) = \dfrac{-4}{1} = -4$, so the curve passes through $(0, -4)$.
- x-intercepts: $f(x) = 0 \iff x^2 - 4 = 0 \iff x = 2$ or $x = -2$; both values are in the domain. The curve cuts the x-axis at $(-2, 0)$ and $(2, 0)$.

WARNING **Do not solve $f(x) = 0$ outside the domain**

A candidate x-intercept must belong to the domain of definition. For $g(x) = \dfrac{x^2 - 1}{x - 1}$, the equation $g(x) = 0$ gives $x = \pm 1$, but $x = 1$ is excluded from the domain: only $(-1, 0)$ is an intercept. (In fact $g(x) = x + 1$ for $x \neq 1$: the curve is a straight line with a "hole" at $x = 1$.)

2.2 Vertical and Horizontal Asymptotes

Imagine driving along the curve of $f(x) = \dfrac{1}{x - 2}$ towards $x = 2$. The road does not stop; it becomes a wall that the curve approaches but never touches. That wall is a **vertical asymptote**.

DEFINITION **Vertical asymptote**

If $\lim_{x \to a^-} f(x) = \pm\infty$ or $\lim_{x \to a^+} f(x) = \pm\infty$, then the line $x = a$ is a **vertical asymptote** to C_f.

In practice, vertical asymptotes occur at values excluded from the domain where the function "blows up" — typically where the denominator of a rational function vanishes while the numerator does not.

DEFINITION **Horizontal asymptote**

If $\lim_{x \to +\infty} f(x) = b$ (a finite number), then the line $y = b$ is a **horizontal asymptote** to C_f at $+\infty$. The same holds at $-\infty$.

EXEMPLE **Vertical and horizontal asymptotes**

Let $f(x) = \dfrac{2x+1}{x-3}$ on $\mathbb{R} \setminus \{3\}$.

- $\lim_{x \to 3^+} f(x) = +\infty$ and $\lim_{x \to 3^-} f(x) = -\infty$: the line $x = 3$ is a vertical asymptote.
- $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \dfrac{2x}{x} = 2$: the line $y = 2$ is a horizontal asymptote at both ends.

WARNING **A horizontal asymptote can be crossed!**

A very common mistake at the GCE is to believe that the curve may never touch its horizontal asymptote. This is false. The asymptote describes the behaviour at infinity, not near the origin. For example, $f(x) = \dfrac{\sin x}{x}$ crosses its horizontal asymptote $y = 0$ infinitely many times, yet $\lim_{x \to +\infty} f(x) = 0$. A curve may cross a horizontal or oblique asymptote — only a vertical asymptote can never be crossed (the function is not defined there).

2.3 Oblique Asymptotes

Some curves, like $f(x) = \dfrac{x^2+1}{x-1}$, have no horizontal asymptote because $f(x) \to \pm\infty$ at infinity — yet they do not wander randomly: they straighten up along a slanting line. Such a line is an **oblique asymptote**.

DEFINITION **Oblique asymptote**

The line $y = mx + c$ (with $m \neq 0$) is an **oblique asymptote** to C_f at $+\infty$ (resp. $-\infty$) if

$$\lim_{x \to +\infty} \left[f(x) - (mx + c) \right] = 0 \quad \text{(resp. at } -\infty\text{)}.$$

How do we find m and c? If $f(x) - (mx + c) \to 0$, then dividing by x (which tends to infinity) gives $\frac{f(x)}{x} - m - \frac{c}{x} \to 0$, hence $\frac{f(x)}{x} \to m$. Once m is known, c is recovered from $f(x) - mx \to c$. This justifies the following method (the derivation above is instructive and may be asked as a justification).

METHOD **Finding an oblique asymptote in two steps**

1. Compute the slope: $m = \lim_{x \to \pm\infty} \frac{f(x)}{x}$. If m is finite and non-zero, continue. If $m = 0$, look for a horizontal asymptote instead; if the limit is infinite, the branch is parabolic (see below).
2. Compute the intercept: $c = \lim_{x \to \pm\infty} \left[f(x) - mx \right]$. If c is finite, the line $y = mx + c$ is an oblique asymptote. If this limit is infinite or fails to exist, there is no oblique asymptote (parabolic branch).

Repeat the study at $+\infty$ and at $-\infty$ separately: the answers may differ.

WARNING **Compute BOTH m and c**

A frequent error is to compute only $m = \lim f(x)/x$, find $m = 1$, and declare "$y = x$ is an asymptote". This is wrong: $f(x) = x + 5 + \frac{1}{x}$ has $\lim f(x)/x = 1$, but its asymptote is $y = x + 5$, not $y = x$. The slope alone never determines the line. Always complete step 2.

WORKED EXAMPLE 5 **Complete asymptote study of a rational function**

Let $f(x) = \frac{x^2 + 1}{x - 1}$ and let C_f be its curve. Determine all asymptotes of C_f.

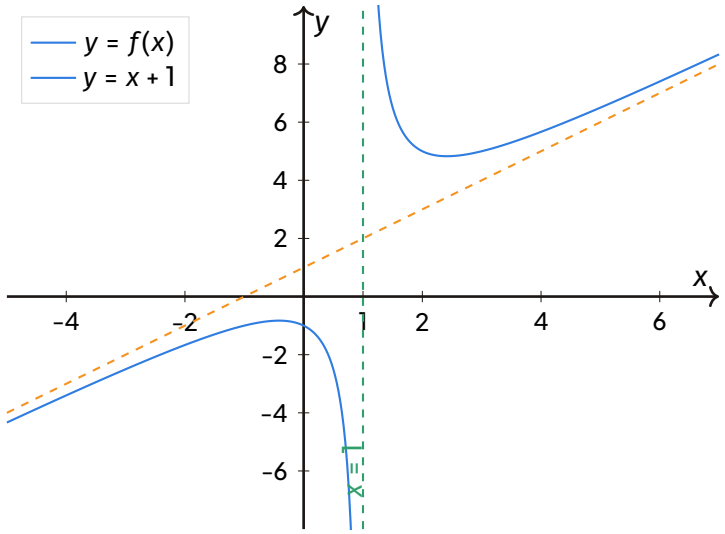
SOLUTION

Domain: f is defined on $\mathbb{R} \setminus \{1\}$.
Vertical asymptote. $\lim_{x \to 1}(x^2 + 1) = 2 > 0$ and $x - 1 \to 0$, so

$$\lim_{x \to 1^+} f(x) = +\infty, \qquad \lim_{x \to 1^-} f(x) = -\infty.$$

The line $x = 1$ is a vertical asymptote.

Behaviour at infinity. $\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} \frac{x^2}{x} = \pm\infty$: no horizontal asymptote. We search for an oblique one.
Step 1: $\frac{f(x)}{x} = \frac{x^2+1}{x(x-1)} = \frac{x^2+1}{x^2-x} \xrightarrow[x\to\pm\infty]{} 1$, so $m = 1$.
Step 2: $f(x) - x = \frac{x^2+1-x(x-1)}{x-1} = \frac{x+1}{x-1} \xrightarrow[x\to\pm\infty]{} 1$, so $c = 1$.
Conclusion: the line $y = x + 1$ is an oblique asymptote at both $+\infty$ and $-\infty$. (Check: polynomial division gives $f(x) = x+1+\frac{2}{x-1}$, and $\frac{2}{x-1} \to 0$ — a fast alternative for rational functions, which also gives the position of the curve relative to the asymptote for free.)



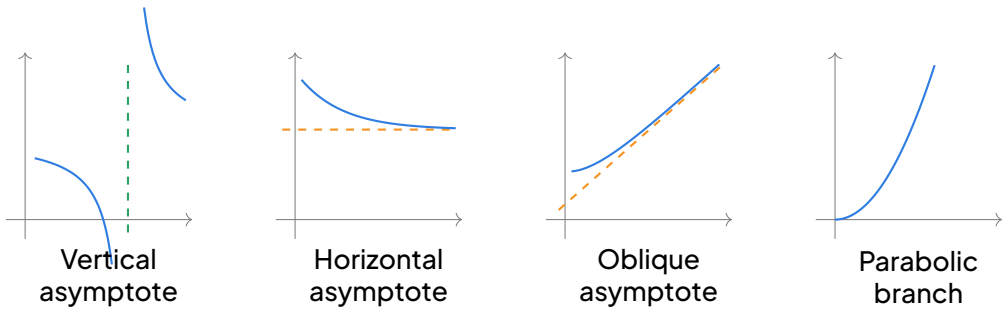
The curve of $f(x) = \frac{x^2+1}{x-1}$ approaching its vertical asymptote $x = 1$ and its oblique asymptote $y = x + 1$.

2.4 Infinite Branches: the Complete Classification

An **infinite branch** is a part of the curve that escapes to infinity. Classifying it means saying how it escapes. Suppose $\lim_{x\to+\infty} f(x) = \pm\infty$ (otherwise we have a horizontal asymptote). We examine $\lim_{x\to+\infty} \frac{f(x)}{x}$:

Limits	Conclusion	Example
$\lim f(x) = b$ (finite)	Horizontal asymptote $y = b$	$f(x) = \dfrac{2x+1}{x-3}$
$\lim f(x) = \pm\infty, \lim \dfrac{f(x)}{x} = m \neq 0, \lim[f(x) - mx] = c$	Oblique asymptote $y = mx + c$	$f(x) = \dfrac{x^2+1}{x-1}$
$\lim f(x) = \pm\infty, \lim \dfrac{f(x)}{x} = \pm\infty$	**Parabolic branch** in the direction of the y-axis	$f(x) = x^2$
$\lim f(x) = \pm\infty, \lim \dfrac{f(x)}{x} = 0$	**Parabolic branch** in the direction of the x-axis	$f(x) = \sqrt{x}$

The fourth case of an infinite branch is the vertical asymptote itself (infinite limit at a finite point). A **parabolic branch** means the curve bends away like a parabola: either it grows faster than any line ($f(x)/x \to \infty$, vertical direction, like x^2), or it grows slower than any line ($f(x)/x \to 0$, horizontal direction, like $\sqrt{x}$). In both cases the search for an oblique asymptote fails, and that failure is precisely the conclusion to state in the examination.



The four types of infinite branches: vertical asymptote, horizontal asymptote, oblique asymptote, parabolic branch.

2.5 Position of a Curve with Respect to a Line

Knowing the asymptote is only half the story: the examiner will also ask on which side of it the curve lies. The idea is beautifully simple. Two points with the same abscissa x, one on the curve $(x, f(x))$ and one on the line $(x, g(x))$, are compared by the sign of the **difference function** $d(x) = f(x) - g(x)$.

PROPERTY **Position of C_f relative to a line**

Let C_f be the curve of f and Δ the line $y = g(x) = mx + c$, and let $d(x) = f(x) - g(x)$ on a common

interval.

- If $d(x) > 0$, then C_f is above Δ.
- If $d(x) < 0$, then C_f is below Δ.
- If $d(x) = 0$, the curve and the line meet at $(x, f(x))$.

This property is applied directly; its justification is the definition of the order on the ordinates of points with the same abscissa.

PROPERTY **Crossing an asymptote**

A curve crosses its horizontal or oblique asymptote $y = mx + c$ exactly at the points whose abscissae solve $f(x) = mx + c$, i.e. $d(x) = 0$. If the sign of d changes there, the curve passes from one side of the asymptote to the other.

METHOD **Studying the position of a curve relative to a line**

1. Form the difference $d(x) = f(x) - g(x)$ and reduce it to a single fraction.
2. Factorise numerator and denominator as much as possible.
3. Build a sign table of $d(x)$ on the domain of f (do not forget forbidden values).
4. Conclude in words: “above on …”, “below on …”, “intersects the line at …”.

WORKED EXAMPLE 6 **Position of a curve relative to its oblique asymptote**

Let $f(x) = \dfrac{x^2+1}{x-1}$ with oblique asymptote $\Delta : y = x + 1$. Study the position of C_f relative to Δ.

SOLUTION

We compute the difference:

$$d(x) = f(x) - (x+1) = \frac{x^2+1}{x-1} - (x+1) = \frac{x^2+1-(x+1)(x-1)}{x-1} = \frac{x^2+1-x^2+1}{x-1} = \frac{2}{x-1}.$$

The numerator 2 is positive, so $d(x)$ has the sign of $x - 1$:

x	$-\infty$		1		$+\infty$
$d(x) = \dfrac{2}{x-1}$		$-$	‖	$+$	

Conclusion:

- On $]-\infty, 1[$, $d(x) < 0$: C_f is below Δ.
- On $]1, +\infty[$, $d(x) > 0$: C_f is above Δ.
- $d(x) = 0$ has no solution: the curve never crosses its oblique asymptote.

This matches the figure above: the left branch creeps up under the line $y = x + 1$, the right branch descends onto it from above.

WORKED EXAMPLE 7 **Position relative to a given line, with crossing**

Let $f(x) = \dfrac{x^2 - 3x + 3}{x - 2}$ on $\mathbb{R} \setminus \{2\}$. Show that $\Delta : y = x - 1$ is an oblique asymptote and study the position of C_f relative to Δ.

SOLUTION

Asymptote. $\dfrac{f(x)}{x} = \dfrac{x^2 - 3x + 3}{x^2 - 2x} \to 1 = m$; then

$$f(x) - x = \frac{x^2 - 3x + 3 - x(x - 2)}{x - 2} = \frac{-x + 3}{x - 2} \to -1 = c.$$

So $y = x - 1$ is an oblique asymptote at $\pm\infty$.

Position. $d(x) = f(x) - (x - 1) = \dfrac{x^2 - 3x + 3 - (x - 1)(x - 2)}{x - 2} = \dfrac{1}{x - 2}$.

x	$-\infty$		2		$+\infty$
$d(x)$		$-$	$\|\|$	$+$	

C_f is below Δ on $]-\infty, 2[$ and above on $]2, +\infty[$; since d never vanishes, there is no intersection.

A case with crossing. Take $h(x) = \dfrac{x^2 + 4x + 2}{x + 1}$ on $\mathbb{R} \setminus \{-1\}$. Polynomial division gives $h(x) = x + 2 + \dfrac{x}{x + 1}$, so $y = x + 2$ is the oblique asymptote and $d(x) = \dfrac{x}{x + 1}$, which vanishes at $x = 0$: the curve crosses its asymptote at the point $(0, 2)$, passing from above (for $-1 < x < 0$, where $d > 0$) to below (for $x > 0$, where $d < 0$).

WARNING **Sign table must respect the domain**

When making the sign table of $d(x)$, always include the forbidden values of f (double bar $\|\|$). The position of the curve can change across a vertical asymptote even when d does not vanish there, exactly as in the two examples above.

KEY POINTS **Key points of the section**

- Intercepts: compute $f(0)$ and solve $f(x) = 0$ within the domain.
- $\lim_{x\to a} f(x) = \pm\infty \Rightarrow$ vertical asymptote $x = a$; $\lim_{x\to\pm\infty} f(x) = b \Rightarrow$ horizontal asymptote $y = b$.
- Oblique asymptote: $m = \lim \dfrac{f(x)}{x}$, then $c = \lim[f(x) - mx]$; both must be finite, with $m \neq 0$.
- If the oblique search fails ($f(x)/x \to \infty$ or $\to 0$), the branch is parabolic (vertical or horizontal direction).
- Position relative to a line: sign of $d(x) = f(x) - g(x)$; the curve crosses its horizontal/oblique asymptote where $d(x) = 0$.
- A horizontal or oblique asymptote may be crossed; a vertical one never.

EXERCISE 2 **Asymptotes and position**

Let $f(x) = \dfrac{x^2 + 2x + 2}{x + 1}$ and C_f its curve.

1. Give the domain of f and show that $x = -1$ is a vertical asymptote.
2. Show that the line $y = x + 1$ is an oblique asymptote at $+\infty$ and at $-\infty$.
3. Study the position of C_f relative to the line $y = x + 1$.
4. Find the intercepts of C_f with the axes.

SOLUTION **Exercise 1**

1. Domain: $\mathbb{R} \setminus \{-1\}$. $\lim_{x\to -1}(x^2+2x+2) = 1 > 0$ and $x+1 \to 0$, so $\lim_{x\to -1^+} f = +\infty$, $\lim_{x\to -1^-} f = -\infty$: $x = -1$ is a vertical asymptote.

2. $\dfrac{f(x)}{x} = \dfrac{x^2 + 2x + 2}{x^2 + x} \to 1 = m$. Then $f(x) - x = \dfrac{x^2 + 2x + 2 - x^2 - x}{x + 1} = \dfrac{x + 2}{x + 1} \to 1 = c$. Hence $y = x + 1$ is an oblique asymptote at both ends.

3. $d(x) = f(x) - (x + 1) = \dfrac{x^2 + 2x + 2 - (x + 1)^2}{x + 1} = \dfrac{1}{x + 1}$. So $d(x) < 0$ on $]-\infty, -1[$ (curve below the line) and $d(x) > 0$ on $]-1, +\infty[$ (curve above the line); d never vanishes, so no crossing.

4. $f(0) = 2$: y-intercept $(0, 2)$. $f(x) = 0 \iff x^2 + 2x + 2 = 0$, discriminant $4 - 8 = -4 < 0$: no x-intercept.

DID YOU KNOW? **Asymptotes in real life**

Asymptotes are not only exam material. The time needed to transmit data over a network as more users connect, the concentration of a medicine in the blood after an injection, or the cost per unit in a factory in Douala as production grows — all these quantities approach horizontal asymptotes. Recognising an asymptote means understanding the long-term behaviour of a real system.

3 STATIONARY POINTS AND VARIATION OF A FUNCTION

In the previous section we learned how to describe the global behaviour of a curve: where it meets the axes, how it behaves at infinity, and which lines it approaches. We now turn to the local behaviour: where does the curve rise, where does it fall, and where does it turn around? The tool that answers all these questions is the **derivative**, and the document that summarises all the answers is the **table of variation**. Mastering this table is essential: at the GCE Advanced Level, almost every curve-sketching question requires it, and it is also the key to counting the solutions of an equation $f(x) = k$.

3.1 Derivative and Monotonicity

Intuition.

Imagine driving from Bamenda to Bafoussam along a hilly road, and think of the road profile as the graph of a function f. Where the car climbs, the slope is positive; where it descends, the slope is negative. The slope of the road at a point is exactly the slope of the tangent line to the graph, that is, the number $f'(x)$. It is therefore natural that the sign of f' should tell us whether f is increasing or decreasing.

PROPERTY **Derivative and monotonicity**

Let f be differentiable on an interval I.

- If $f'(x) > 0$ for all $x \in I$ (except possibly at isolated points where $f' = 0$), then f is **strictly increasing** on I.
- If $f'(x) < 0$ for all $x \in I$ (except possibly at isolated points where $f' = 0$), then f is **strictly decreasing** on I.
- If $f'(x) = 0$ for all $x \in I$, then f is **constant** on I.

Why this is true.

The rigorous proof uses the Mean Value Theorem (which is part of the Further Mathematics syllabus). The idea is simple: if $f'(x) > 0$ on I, then for any $a < b$ in I the chord joining $(a, f(a))$ to $(b, f(b))$ has gradient $\dfrac{f(b) - f(a)}{b - a}$. The Mean Value Theorem guarantees a point $c \in (a, b)$ where

the tangent is parallel to this chord: $f'(c) = \dfrac{f(b) - f(a)}{b - a} > 0$. Since $b - a > 0$, we get $f(b) - f(a) > 0$, i.e. $f(b) > f(a)$: the function increases.

WARNING **Sign of f' versus sign of f**

Do not confuse the sign of $f'(x)$ with the sign of $f(x)$. The sign of f tells you whether the curve is above or below the x-axis; the sign of f' tells you whether the curve is rising or falling. A function can be negative and increasing (for example $f(x) = x$ for $x < 0$).

3.2 Stationary Points and Their Nature

Intuition.

At the very top of a hill and at the very bottom of a valley, the road is momentarily flat: the tangent to the graph is horizontal, so $f'(x) = 0$. Such points deserve a name.

DEFINITION **Stationary point**

A **stationary point** of a differentiable function f is a point a of its domain such that

$$f'(a) = 0.$$

Geometrically, the tangent to the curve at $(a, f(a))$ is horizontal.

A stationary point is of one of three types:

- a **local maximum**: $f(a) \geq f(x)$ for all x near a (the curve turns from rising to falling);
- a **local minimum**: $f(a) \leq f(x)$ for all x near a (the curve turns from falling to rising);
- a **horizontal point of inflexion**: the tangent is horizontal but the curve does not turn around; it keeps rising (or keeps falling), as for $f(x) = x^3$ at $x = 0$.

WARNING **$f'(a) = 0$ does not guarantee an extremum**

The most frequent mistake at the GCE is to write "$f'(a) = 0$, therefore f has a maximum (or minimum) at a". This is false in general: for $f(x) = x^3$ we have $f'(0) = 0$, yet f is strictly increasing on $\mathbb{R}$ and has no extremum at 0. The condition $f'(a) = 0$ is necessary but not sufficient. You must always check the sign change of f' around a (first derivative test) or use the second derivative test.

PROPERTY **First derivative test**

Let f be differentiable on an open interval containing a, with $f'(a) = 0$.

- If f' changes sign from + to − at a, then f has a **local maximum** at a.
- If f' changes sign from − to + at a, then f has a **local minimum** at a.
- If f' keeps the same sign on both sides of a, then a is a **horizontal point of inflexion**.

Justification. If $f' > 0$ just before a and $f' < 0$ just after, then f is increasing before a and decreasing after a, so $f(x) \leq f(a)$ on each side: $f(a)$ is a local maximum. The other cases are similar. This justification is examinable as a short reasoning question.

PROPERTY **Second derivative test**

Let f be twice differentiable near a, with $f'(a) = 0$.

- If $f''(a) > 0$, then f has a **local minimum** at a.
- If $f''(a) < 0$, then f has a **local maximum** at a.
- If $f''(a) = 0$, the test is inconclusive: use the first derivative test.

Idea of the proof. If $f''(a) > 0$, then f' is increasing through a; since $f'(a) = 0$, f' must be negative just before a and positive just after, which is exactly the minimum case of the first derivative test.

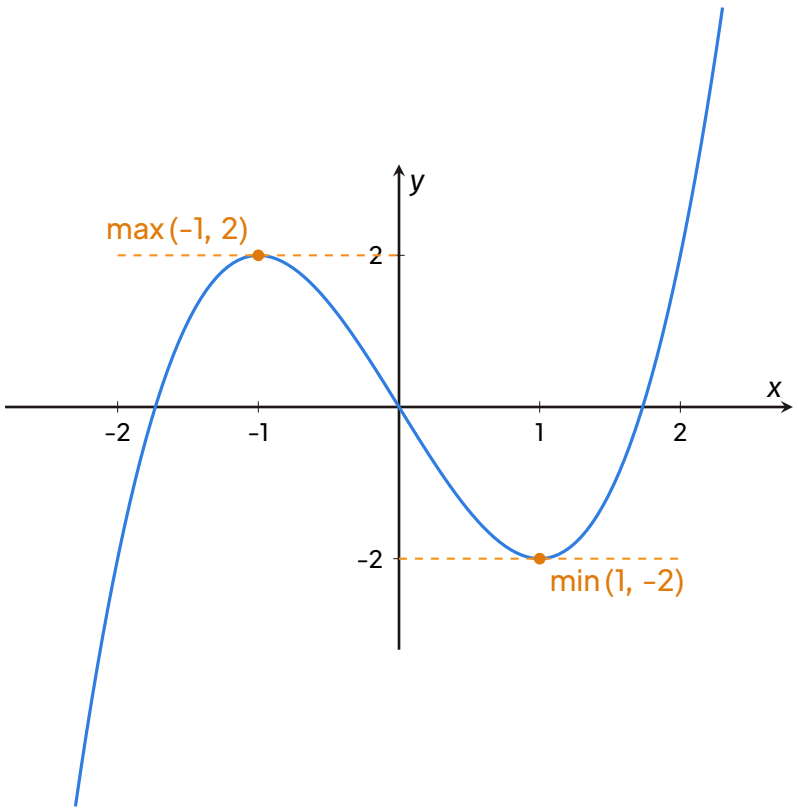
WARNING **The inconclusive case**

$f''(a) = 0$ does not mean "no extremum". Compare $f(x) = x^4$ and $g(x) = x^3$ at $a = 0$: both have $f'(0) = f''(0) = 0$, but x^4 has a minimum at 0 while x^3 has a horizontal inflexion. When $f''(a) = 0$, go back to the sign of f'.

EXEMPLE **Classifying stationary points**

Let $f(x) = x^3 - 3x$. Then $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$, so the stationary points are $x = -1$ and $x = 1$. Also $f''(x) = 6x$.

- At $x = -1$: $f''(-1) = -6 < 0$, so f has a local maximum $f(-1) = 2$.
- At $x = 1$: $f''(1) = 6 > 0$, so f has a local minimum $f(1) = -2$.



The curve of $f(x) = x^3 - 3x$ with horizontal tangents (dashed) at its stationary points.

3.3 Sign Table of $f'(x)$ and Construction of the Table of Variation

To exploit the link between the sign of f' and the monotonicity of f, we first study the sign of f' on the whole domain. The strategy is always the same: **factorise** $f'(x)$ as much as possible, find its zeros (the **critical values**), and use the sign of each factor.

METHOD **Building a table of variation in four steps**

1. Domain. Determine the domain of definition D_f and compute the limits of f at the bounds of D_f (open bounds, vertical asymptotes, $\pm\infty$).
2. Derivative. Compute $f'(x)$ and factorise it completely; solve $f'(x) = 0$ to find the critical values, and note any value where f' does not exist.
3. Sign. Build the sign table of $f'(x)$ over D_f, marking every critical value and every excluded value with the appropriate symbol (0 or double bar).
4. Limits and extrema. Draw the arrows of variation (↗ where $f' > 0$, ↘ where $f' < 0$) and write the value of f at each remarkable point: limits at the bounds and exact values $f(a)$ at the stationary points.

WORKED EXAMPLE 8 **Complete study of $f(x) = x^3 - 3x$**

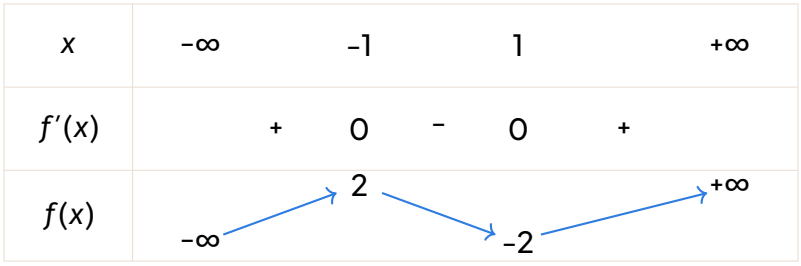
Study the variations of $f(x) = x^3 - 3x$ on $\mathbb{R}$ and draw its table of variation.

SOLUTION

Step 1 — Domain and limits. f is a polynomial, so $D_f = \mathbb{R}$. As $x \to +\infty$, x^3 dominates, so $\lim_{+\infty} f = +\infty$; as $x \to -\infty$, $\lim_{-\infty} f = -\infty$.
Step 2 — Derivative. $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$; critical values $x = -1$ and $x = 1$.
Step 3 — Sign of f'. The quadratic $3(x - 1)(x + 1)$ is positive outside its roots and negative between them: $f' > 0$ on $(-\infty, -1) \cup (1, +\infty)$ and $f' < 0$ on $(-1, 1)$.
Step 4 — Values. $f(-1) = -1 + 3 = 2$ (local maximum), $f(1) = 1 - 3 = -2$ (local minimum). Hence the table:

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

The curve rises from $-\infty$ to the peak $(-1, 2)$, falls to the valley $(1, -2)$, then rises to $+\infty$.



Template of the table of variation of $f(x) = x^3 - 3x$.

WARNING **Values where f is not defined**

Never write a finite value of f in the table at a point where f is not defined. For example, for $g(x) = \dfrac{x^2}{x - 1}$, the value $x = 1$ is excluded from the domain: in the table you must place a double vertical bar at $x = 1$ in every row and write the limits $\lim_{x \to 1^-} g(x) = -\infty$ and $\lim_{x \to 1^+} g(x) = +\infty$ on either side — never $g(1)$. A value written under a double bar is a serious error that examiners penalise immediately.

3.4 Reading Information from the Table of Variation

A correct table of variation is a goldmine of information. From it we read directly:

- the **extrema** of f (local maxima and minima, with their values);
- the **range** of f on each monotonic branch;
- the **number of solutions** of the equation $f(x) = k$ for any constant k.

PROPERTY **Counting the solutions of $f(x) = k$**

On each interval where f is continuous and strictly monotonic, the equation $f(x) = k$ has exactly one solution if k lies strictly between the values (or limits) of f at the ends of that interval, and no solution otherwise. This follows from the Intermediate Value Theorem (existence) together with strict monotonicity (uniqueness). The total number of solutions is the sum over all monotonic branches, and it is read directly from the table of variation.

EXEMPLE **Counting solutions from the table**

For $f(x) = x^3 - 3x$, the table above shows three monotonic branches:

- on $(-\infty, -1]$, f goes from $-\infty$ to 2: one solution of $f(x) = k$ iff $k \le 2$;
- on $[-1, 1]$, f goes from 2 down to -2: one solution iff $-2 \le k \le 2$;
- on $[1, +\infty)$, f goes from -2 to $+\infty$: one solution iff $k \ge -2$.

Counting carefully (and not double-counting the shared endpoints): if $|k| < 2$ the equation has three solutions; if $k = \pm 2$ it has two solutions (one of them a double root at the extremum); if $|k| > 2$ it has one solution. This is exactly the kind of discussion the GCE Board asks for.

3.5 Interpretation of the Sign of $f(x)$: Position of the Curve Relative to the x-Axis

The table of variation also tells us the sign of f itself, which locates the curve relative to the x-axis. For a point (x, y) in the plane, the sign of $y - f(x)$ tells us whether the point lies above ($y - f(x) > 0$) or below ($y - f(x) < 0$) the curve $y = f(x)$. In particular, on the curve itself $y = f(x)$ so $y - f(x) = 0$.

- if $f(x) > 0$ at x, the curve lies **above** the x-axis at that abscissa;
- if $f(x) < 0$, the curve lies **below** the x-axis;
- the zeros of f are the abscissae of the **intercepts** with the x-axis.

More generally, comparing the curve with a line $y = mx + c$ amounts to studying the sign of the difference $g(x) = f(x) - (mx + c)$: the curve is above the line where $g(x) > 0$ and below it where $g(x) < 0$. This idea will be developed fully when we study the position of a curve with respect to a line.

WORKED EXAMPLE 9 **Sign of f from the table of variation**

Using the table of variation of $f(x) = x^3 - 3x$, determine the sign of $f(x)$ on $\mathbb{R}$.

SOLUTION

Factorise: $f(x) = x(x^2 - 3) = x(x - \sqrt{3})(x + \sqrt{3})$, so the zeros are $-\sqrt{3}$, 0, $\sqrt{3}$. From the table: on $(-\infty, -1]$, f increases from $-\infty$ to 2, so it crosses zero exactly once, at $x = -\sqrt{3}$; hence $f < 0$ on $(-\infty, -\sqrt{3})$ and $f > 0$ on $(-\sqrt{3}, -1]$. On $[-1, 1]$, f decreases from 2 to -2, crossing zero once, at $x = 0$: $f > 0$ on $[-1, 0)$ and $f < 0$ on $(0, 1]$. On $[1, +\infty)$, f increases from -2 to $+\infty$, crossing zero once, at $x = \sqrt{3}$: $f < 0$ on $[1, \sqrt{3})$ and $f > 0$ on $(\sqrt{3}, +\infty)$. Conclusion: the curve is above the x-axis on $(-\sqrt{3}, 0) \cup (\sqrt{3}, +\infty)$ and below it on $(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$.

EXERCISE 3 **Practice**

Let $f(x) = x^3 - 3x^2 + 2$ defined on $\mathbb{R}$.

1. Compute $f'(x)$, factorise it, and find the stationary points.
2. Use the second derivative test to determine their nature.
3. Draw the complete table of variation of f.
4. Deduce the number of solutions of $f(x) = k$ according to the values of k.

SOLUTION **Exercise 1**

1. $f'(x) = 3x^2 - 6x = 3x(x - 2)$; stationary points $x = 0$ and $x = 2$.
2. $f''(x) = 6x - 6$. $f''(0) = -6 < 0$: local maximum $f(0) = 2$. $f''(2) = 6 > 0$: local minimum $f(2) = 8 - 12 + 2 = -2$.
3. Limits: $\lim_{-\infty} f = -\infty$, $\lim_{+\infty} f = +\infty$. Sign of f': positive on $(-\infty, 0) \cup (2, +\infty)$, negative on $(0, 2)$.

x	$-\infty$		0		2		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

4. By the Intermediate Value Theorem on each branch: if $k > 2$ or $k < -2$, one solution; if $k = 2$ or $k = -2$, two solutions; if $-2 < k < 2$, three solutions.

KEY POINTS **Key points of this section**

- f increases where $f' > 0$ and decreases where $f' < 0$; the sign of f' governs the shape, the sign of f governs the position relative to the x-axis.
- A stationary point satisfies $f'(a) = 0$ (horizontal tangent); it is a maximum, a minimum, or a horizontal inflexion — decide with the sign change of f' or with $f''(a)$.
- The table of variation is built in four steps: domain and limits, derivative, sign of f', arrows with exact values.
- The number of solutions of $f(x) = k$ is read from the table by applying the Intermediate Value Theorem on each monotonic branch.
- Never place a value of f at a point excluded from the domain: use a double bar and limits.

With the variations of f now under control, one refinement remains before we can sketch curves with full confidence: the way the curve bends. This is the question of concavity and points of inflexion, which we study in the next section.

4 CONCAVITY, POINTS OF INFLEXION AND SYMMETRY

In the previous section we learnt how the first derivative f' reveals where a function is increasing or decreasing and where its stationary points lie. But two curves can both be increasing on an interval and yet look completely different: one may bend upwards like a cup, the other downwards like a cap. To distinguish these shapes we need the second derivative f''. In this section we study the **concavity** of a curve, the **points of inflexion** where the bending changes direction, and the **symmetries** (centre and axis) that allow us to halve our work when sketching a curve. By the end, you will have the complete study plan of a function.

4.1 Concavity of a Curve

Intuition. Imagine driving along the graph of f from left to right. If the road bends so that you keep turning left (anti-clockwise), the curve is **concave up**; if you keep turning right (clockwise), it is **concave down**. Equivalently: a concave-up curve holds water like a bowl; a concave-down curve sheds water like a roof.

Geometrically, concavity is detected by the position of the curve relative to its **tangent lines**:

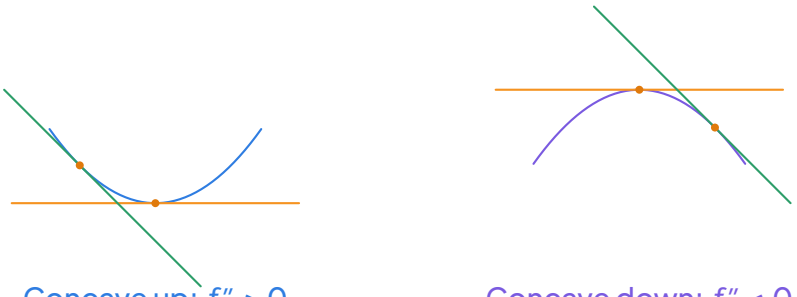
- if near a point the curve lies **above** each of its tangents, it is concave up;
- if near a point the curve lies **below** each of its tangents, it is concave down.

Why does the second derivative decide this? The slope of the tangent is $f'(x)$. If $f''(x) > 0$ on an interval, then f' is increasing: the tangent turns anti-clockwise as x grows, so the curve bends upwards and stays above its tangents. If $f''(x) < 0$, then f' is decreasing and the curve bends downwards, staying below its tangents.

DEFINITION **Concavity**

Let f be twice differentiable on an interval I.

- f is **concave up** (convex) on I if $f''(x) > 0$ for all $x \in I$; the curve lies above each of its tangents on I.
- f is **concave down** (concave) on I if $f''(x) < 0$ for all $x \in I$; the curve lies below each of its tangents on I.



Concavity seen through tangent lines: concave up (left, tangents at $x = 0$ and $x = -1$), concave down (right, tangents at $x = 0$ and $x = 1$).

EXEMPLE **Testing concavity**

Let $f(x) = x^3 - 3x$. Then $f'(x) = 3x^2 - 3$ and $f''(x) = 6x$.

- For $x < 0$, $f''(x) < 0$: the curve is concave down on $]-\infty, 0[$.
- For $x > 0$, $f''(x) > 0$: the curve is concave up on $]0, +\infty[$.

4.2 Points of Inflexion

Intuition. When you drive over the top of a hill and the road changes from bending right to bending left, there is exactly one instant when the steering wheel passes through the neutral position. On a curve, that instant is a **point of inflexion**: the point where the concavity changes.

DEFINITION **Point of inflexion**

A point $M(a, f(a))$ on the curve of f is a **point of inflexion** if the concavity of f changes at a:

the curve crosses its tangent at M, passing from one side of the tangent to the other.

PROPERTY **Condition for a point of inflexion**

Let f be twice differentiable on an open interval containing a.

- Necessary condition: if $(a, f(a))$ is a point of inflexion, then $f''(a) = 0$.
- Sufficient condition: if $f''(a) = 0$ and f'' changes sign at a (i.e. the concavity changes), then $(a, f(a))$ is a point of inflexion.

(The sign change of f'' is the examinable justification; always exhibit it, e.g. in a sign table.)

WARNING $f''(a) = 0$ **is not enough**

A very common mistake at the GCE is to write "$f''(a) = 0$, hence inflexion point". This is false without a sign change. Counter-example: $f(x) = x^4$ has $f''(x) = 12x^2$, so $f''(0) = 0$, yet $f''(x) > 0$ on both sides of 0: the curve stays concave up and $(0, 0)$ is not an inflexion point (it is a minimum). Always verify that f'' changes sign.

Stationary versus ordinary inflexion points. An inflexion point $(a, f(a))$ is called:

- a **stationary inflexion point** if in addition $f'(a) = 0$: the tangent there is horizontal (example: $f(x) = x^3$ at the origin);
- an **ordinary inflexion point** if $f'(a) \neq 0$: the tangent there is oblique (example: $f(x) = x^3 - 3x$ at the origin, where $f'(0) = -3$).

To classify, compute $f'(a)$ after confirming the sign change of f''.

WORKED EXAMPLE 10 **Inflexion points of a cubic**

Statement. Let $f(x) = x^3 - 3x$ defined on $\mathbb{R}$. Determine the concavity of f and find its points of inflexion, specifying their nature.

SOLUTION

Solution. We compute $f'(x) = 3x^2 - 3$ and $f''(x) = 6x$. $f''(x) = 0 \iff x = 0$, and $f''(x) = 6x$ changes sign at 0:

x	$-\infty$		0		$+\infty$
$f''(x)$		$-$	0	$+$	
Concavity		down	infl.	up	

Since f'' changes sign at 0, the point $(0, f(0)) = (0, 0)$ is a point of inflexion. Moreover $f'(0) = -3 \neq 0$, so it is an ordinary inflexion point; the tangent there has equation $y = -3x$, and indeed the curve crosses this line at the origin since $x^3 - 3x - (-3x) = x^3$ changes sign at 0.

4.3 Centre of Symmetry of a Curve

Intuition. Some curves look exactly the same after a half-turn about a special point I. The graph of $y = x^3$ rotated through 180° about the origin falls back on itself. Such a point I is called a **centre of symmetry**. Detecting it is precious: half of the curve determines the other half.

DEFINITION **Centre of symmetry**

The point $I(a, b)$ is a **centre of symmetry** of the curve C_f if for every x such that $a + x$ and $a - x$ are in the domain,

$$f(a + x) + f(a - x) = 2b.$$

Equivalently, with $M(a + x, f(a + x))$ and $M'(a - x, f(a - x))$, the point I is the midpoint of $[MM']$.

METHOD **Proving that $I(a, b)$ is a centre of symmetry**

1. Check the domain is symmetric about a: $a + x \in D_f \iff a - x \in D_f$.
2. Compute $f(a + x)$ and $f(a - x)$ separately.
3. Add them and simplify.
4. If $f(a + x) + f(a - x) = 2b$ for all admissible x, conclude that $I(a, b)$ is a centre of symmetry.

To find a candidate (a, b), look at the domain (e.g. an excluded value a for a rational function) and compute $b = f(a)$ or use limits and asymptotes.

WORKED EXAMPLE 11 **Centre of symmetry of a hyperbola**

Statement. Let $f(x) = \dfrac{2x + 1}{x - 1}$ on $D_f = \mathbb{R} \setminus \{1\}$. Show that $I(1, 2)$ is a centre of symmetry of C_f.

SOLUTION

Solution. The domain is symmetric about 1 since $1 + x \in D_f \iff x \neq 0 \iff 1 - x \in D_f$. For

$x \neq 0$:

$$f(1+x) = \frac{2(1+x)+1}{(1+x)-1} = \frac{2x+3}{x} = 2 + \frac{3}{x},$$
$$f(1-x) = \frac{2(1-x)+1}{(1-x)-1} = \frac{3-2x}{-x} = 2 - \frac{3}{x}.$$

Hence

$$f(1+x) + f(1-x) = \left(2 + \tfrac{3}{x}\right) + \left(2 - \tfrac{3}{x}\right) = 4 = 2 \times 2.$$

Therefore $I(1, 2)$ is a centre of symmetry of C_f. (Notice $a = 1$ is the excluded value and $b = 2$ is the horizontal asymptote: for a homographic function the centre of symmetry is the intersection of the two asymptotes.)

4.4 Axis of Symmetry of a Curve

Intuition. Other curves are their own mirror image in a vertical line, like the parabola $y = x^2$ in the y-axis. Such a line is an **axis of symmetry**.

DEFINITION **Axis of symmetry**

The vertical line $x = a$ is an **axis of symmetry** of C_f if for every x such that $a + x$ and $a - x$ are in the domain,

$$f(a+x) = f(a-x).$$

The points $M(a + x, f(a + x))$ and $M'(a - x, f(a - x))$ are then mirror images in the line $x = a$.

METHOD **Proving that the line $x = a$ is an axis of symmetry**

1. Check the domain is symmetric about a.
2. Compute $f(a + x)$ and $f(a - x)$.
3. Show the two expressions are equal for all admissible x.
4. Conclude that the line $x = a$ is an axis of symmetry.

EXEMPLE **Axis of symmetry of a parabola**

Let $f(x) = x^2 - 4x + 5$ on $\mathbb{R}$. We test the line $x = 2$:

$$f(2+x) = (2+x)^2 - 4(2+x) + 5 = x^2 + 1, \qquad f(2-x) = (2-x)^2 - 4(2-x) + 5 = x^2 + 1.$$

Since $f(2+x) = f(2-x)$ for all x, the line $x = 2$ is an axis of symmetry. (This is the vertical line through the vertex $(2, 1)$.)

WARNING **Centre versus axis: do not confuse them**

A centre of symmetry is a point: the test is $f(a+x)+f(a-x) = 2b$ (a sum). An axis of symmetry is a line $x = a$: the test is $f(a+x) = f(a-x)$ (an equality of values). Mixing the two tests is a classic source of lost marks. Also note: a curve can have both, one, or neither.

4.5 Even and Odd Functions: the Special Cases

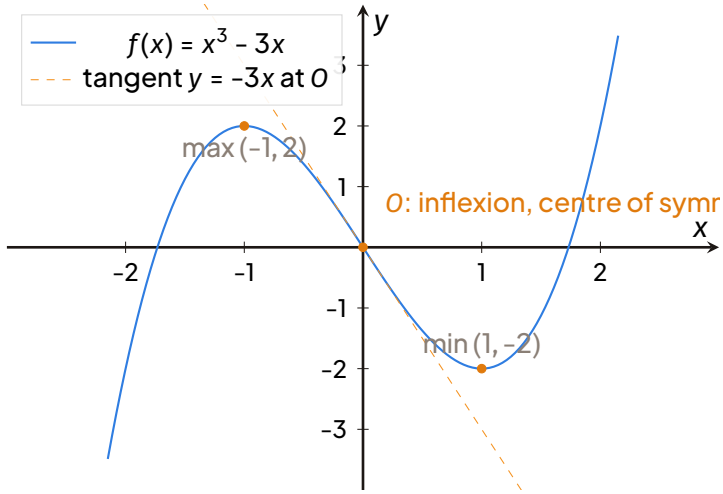
Two symmetries occur so often that they have their own names.

DEFINITION **Even and odd functions**

Let f have a domain symmetric about 0.

- f is **even** if $f(-x) = f(x)$ for all $x \in D_f$: the y-axis (line $x = 0$) is an axis of symmetry.
- f is **odd** if $f(-x) = -f(x)$ for all $x \in D_f$: the origin $O(0, 0)$ is a centre of symmetry.

Examples: x^2, $\cos x$, $|x|$ are even; x^3, $\sin x$, $\frac{1}{x}$ are odd. The function $f(x) = x^3 - 3x$ is odd since $f(-x) = -x^3 + 3x = -f(x)$: its curve has the origin as centre of symmetry.



The curve of $f(x) = x^3 - 3x$: the origin is both a point of inflexion (the curve crosses its tangent $y = -3x$) and a centre of symmetry (f is odd).

DID YOU KNOW? **Why symmetry saves you half the work**

If $I(a, b)$ is a centre of symmetry or $x = a$ is an axis of symmetry, it is enough to study f on half the domain (say $x \geq a$): variations, limits and asymptotes on the other half follow by symmetry. For even or odd functions, study on $[0, +\infty[$ only. GCE examiners reward candidates who state explicitly: "we restrict the study to … and complete by symmetry".

4.6 Assembling Everything: the Complete Study Plan

Concavity, inflexion and symmetry are the final bricks of the full study of a function. Here is the master plan you should follow in every curve-sketching problem.

METHOD **Complete study plan of a function**

1. Domain D_f; reduce it using parity / symmetry (test $f(-x)$, or $f(a \pm x)$).
2. Limits at the bounds of D_f; deduce asymptotes (vertical, horizontal, oblique) and study infinite branches.
3. Intercepts with the axes.
4. First derivative f': sign, stationary points, table of variation.
5. Second derivative f'': sign, concavity, points of inflexion (justify by the sign change of f'').
6. Position of the curve relative to remarkable lines (asymptotes, tangents at inflexion points).
7. Sketch the curve, using symmetry to complete it.

WORKED EXAMPLE 12 **Concavity and symmetry of a rational function**

Statement. Let $g(x) = \dfrac{x^2}{x-1}$ on $\mathbb{R} \setminus \{1\}$. Study the concavity of g and show that $I(1, 2)$ is a centre of symmetry.

SOLUTION

Solution. Concavity. Write $g(x) = x + 1 + \dfrac{1}{x-1}$ (polynomial division). Then

$$g'(x) = 1 - \frac{1}{(x-1)^2}, \qquad g''(x) = \frac{2}{(x-1)^3}.$$

$g''(x)$ has the sign of $x - 1$: negative on $]-\infty, 1[$ (concave down), positive on $]1, +\infty[$ (con-

cave up). Since g is not defined at 1, there is no point of inflexion — the concavity changes across the vertical asymptote $x = 1$, not across a point of the curve.
Centre of symmetry. For $x \neq 0$:

$$g(1+x) = (1+x) + 1 + \frac{1}{x} = 2 + x + \frac{1}{x},$$
$$g(1-x) = (1-x) + 1 - \frac{1}{x} = 2 - x - \frac{1}{x}.$$

Thus $g(1+x)+g(1-x) = 4 = 2\times 2$, so $I(1, 2)$ is a centre of symmetry. Note that I is the intersection of the vertical asymptote $x = 1$ and the oblique asymptote $y = x + 1$.

EXERCISE 4

Let $f(x) = \frac{x^3}{3} - x + 1$ on $\mathbb{R}$.

1. Compute f'' and study the concavity of f.
2. Show that $(0, 1)$ is a point of inflexion and state whether it is stationary or ordinary.
3. Prove that $I(0, 1)$ is a centre of symmetry of C_f.

SOLUTION **Exercise 1**

1. $f'(x) = x^2 - 1$ and $f''(x) = 2x$. Hence $f''(x) < 0$ on $]-\infty, 0[$ (concave down) and $f''(x) > 0$ on $]0, +\infty[$ (concave up).
2. $f''(0) = 0$ and f'' changes sign at 0, so $(0, f(0)) = (0, 1)$ is a point of inflexion. Since $f'(0) = -1 \neq 0$, it is an ordinary inflexion point (tangent of slope -1).
3. Compute:

$$f(x) + f(-x) = \left(\tfrac{x^3}{3} - x + 1\right) + \left(-\tfrac{x^3}{3} + x + 1\right) = 2 = 2 \times 1,$$

so $f(0 + x) + f(0 - x) = 2 \times 1$ for all x: $I(0, 1)$ is a centre of symmetry.

EXERCISE 5

Let $h(x) = \frac{x^2 - 2x + 2}{x - 1}$ on $\mathbb{R} \setminus \{1\}$.

1. Show that $h(x) = x - 1 + \frac{1}{x - 1}$.
2. Deduce that $I(1, 0)$ is a centre of symmetry of C_h.
3. Compute h'' and study the concavity of h. Does h have a point of inflexion?

SOLUTION **Exercise 2**

1. $x - 1 + \dfrac{1}{x-1} = \dfrac{(x-1)^2+1}{x-1} = \dfrac{x^2-2x+2}{x-1} = h(x)$.
2. For $x \neq 0$: $h(1+x) = x + \dfrac{1}{x}$ and $h(1-x) = -x - \dfrac{1}{x}$, so $h(1+x) + h(1-x) = 0 = 2 \times 0$: $I(1, 0)$ is a centre of symmetry.
3. $h'(x) = 1 - \dfrac{1}{(x-1)^2}$, $h''(x) = \dfrac{2}{(x-1)^3}$. Thus h is concave down on $]-\infty, 1[$ and concave up on $]1, +\infty[$. There is no point of inflexion because h is not defined at 1 (the concavity changes across the vertical asymptote).

KEY POINTS **Key points of this section**

- $f'' > 0$: concave up, curve above its tangents; $f'' < 0$: concave down, curve below its tangents.
- Inflexion point at a: requires $f''(a) = 0$ and a sign change of f'' at a. It is stationary if also $f'(a) = 0$, ordinary otherwise.
- Centre of symmetry $I(a, b)$: prove $f(a+x) + f(a-x) = 2b$.
- Axis of symmetry $x = a$: prove $f(a+x) = f(a-x)$.
- Even function ⇒ y-axis symmetry; odd function ⇒ origin symmetry.
- Symmetry halves the study domain; concavity completes the table of variation and refines the sketch.

5 ALGEBRAIC SOLUTIONS OF ABSOLUTE-VALUE INEQUALITIES

In the previous sections we learned to read the shape of a curve: its concavity, its points of inflexion and its possible symmetries. We now change our point of view. Instead of asking "what does the graph look like?", we ask "for which values of x does a given condition hold?" When the condition involves an absolute value, the question becomes an **absolute-value inequality**, and the GCE examiner expects you to solve it algebraically, with complete rigour, and to present the answer as a union of intervals. This section gives you every tool you need.

5.1 Reminder: the Absolute Value and its Properties

Imagine a trader walking along the Douala–Yaoundé highway represented as a number line. Whether she is 5 km to the east or 5 km to the west of a toll gate, her distance from the gate is the same: 5 km. The absolute value is precisely this idea of distance, stripped of direction.

DEFINITION **Absolute value**

For any real number x, the **absolute value** (or **modulus**) of x is

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Geometrically, $|x|$ is the distance from x to 0 on the number line, and $|a - b|$ is the distance between the numbers a and b.

PROPERTY **Main properties of the absolute value**

For all real numbers x and y:

- $|x| \geq 0$, and $|x| = 0 \iff x = 0$;
- $|-x| = |x|$ and $|x|^2 = x^2$;
- $|xy| = |x|\,|y|$ and, for $y \neq 0$, $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$;
- $|x + y| \leq |x| + |y|$ (the **triangle inequality**);
- $\sqrt{x^2} = |x|$ (not x!).

These properties are not usually examined as proofs, but they are used constantly; the last one is a favourite GCE trap.

From the "distance" picture, three basic facts follow immediately for $a > 0$:

$$|x| = a \iff x = a \text{ or } x = -a; \qquad |x| < a \iff -a < x < a; \qquad |x| > a \iff x < -a \text{ or } x > a.$$

In interval notation: $|x| < a$ gives $x \in]-a\,;\,a[$, while $|x| > a$ gives $x \in]-\infty\,;\,-a[\cup]a\,;\,+\infty[$. Notice the union: the solution of a "greater than" modulus inequality is **two separate pieces**, never one interval.

5.2 The Key Equivalences

Everything in this section rests on one idea: replace the modulus by an equivalent condition without modulus.

PROPERTY **Fundamental equivalences**

Let A be any real expression and $b > 0$. Then

$$|A| < b \iff -b < A < b, \qquad |A| > b \iff A < -b \text{ or } A > b.$$

The same holds with $\leq$ and $\geq$ in place of $<$ and $>$.

Justification. $|A|$ is the distance from A to 0. Saying this distance is less than b means A lies strictly between $-b$ and b, i.e. $-b < A < b$. Saying the distance is greater than b means A lies outside that segment, on either side: $A < -b$ or $A > b$. ■

WARNING **A union, not a single interval**

A very common mistake is to write the solution of $|A| > b$ as "$-b > A > b$" or as one interval. This is meaningless: no number is simultaneously less than $-b$ and greater than b. The word is or, and the answer is a union of two intervals.

5.3 Solving $|ax + b| < c$ and $|ax + b| > c$

METHOD **One-modulus inequalities**

1. Check the right-hand side c. If $c \leq 0$, then $|A| < c$ has no solution and $|A| > c$ (with $c < 0$) holds for all x in the domain.
2. If $c > 0$, apply the fundamental equivalence to remove the modulus.
3. Solve the resulting double inequality (or the two separate inequalities) for x, isolating x in the middle.
4. Write the solution set in interval notation and, if useful, draw it on a number line.

WORKED EXAMPLE 13 **Solving $|2x - 3| < 5$**

Solve in $\mathbb{R}$ the inequality $|2x - 3| < 5$ and illustrate the solution on a number line.

SOLUTION

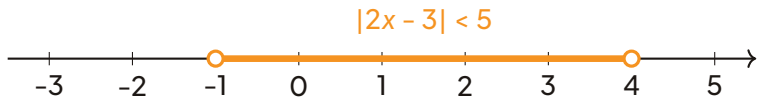
Solution. Since $5 > 0$, the fundamental equivalence gives

$$|2x - 3| < 5 \iff -5 < 2x - 3 < 5.$$

Add 3 to all three parts: $-2 < 2x < 8$. Divide by 2 (positive, so the inequalities keep their direction):

$$-1 < x < 4.$$

Hence $S =]-1; 4[$. Check: $x = 0$ gives $|-3| = 3 < 5$ ✓; $x = 5$ gives $|7| = 7 \not< 5$ ✓ (correctly excluded).



Solution set of $|2x - 3| < 5$: the open interval $]-1; 4[$ (open dots: endpoints excluded).

EXEMPLE **A "greater than" case**

Solve $|3x + 1| \geq 4$. Since $4 > 0$:

$$3x + 1 \leq -4 \quad \text{or} \quad 3x + 1 \geq 4 \iff x \leq -\tfrac{5}{3} \quad \text{or} \quad x \geq 1.$$

So $S = \left]-\infty ; -\tfrac{5}{3}\right] \cup [1 ; +\infty[$. Note the closed brackets because the inequality is non-strict.

5.4 Inequalities with Two Moduli: Squaring Both Sides

How do we solve $|x-3| < |2x+1|$? Neither fundamental equivalence applies directly. The rescue comes from a simple observation: **both sides of $|A| < |B|$ are non-negative**, and the squaring function $t \mapsto t^2$ is strictly increasing on $[0 ; +\infty[$. Therefore squaring preserves the inequality.

PROPERTY **Squaring a two-modulus inequality**

For any real expressions A and B,

$$|A| < |B| \iff A^2 < B^2 \iff (A - B)(A + B) < 0,$$

because $|A|^2 = A^2$ and both sides are non-negative. The same holds with $\leq$, $>$, $\geq$. This derivation is examinable: quote it before using it.

METHOD **Solving $|A| < |B|$**

1. Square both sides (legal, since both are ≥ 0): $A^2 < B^2$.
2. Move everything to one side and factorise as a difference of squares: $(A - B)(A + B) < 0$.
3. Solve the resulting polynomial inequality using a sign table.
4. Write the solution in interval notation and verify with a test value.

WORKED EXAMPLE 14 **Solving $|x-3| < |2x+1|$**

Solve in $\mathbb{R}$: $|x-3| < |2x+1|$.

SOLUTION

Solution. Both sides are non-negative, so we may square:

$$(x-3)^2 < (2x+1)^2 \iff (x-3)^2 - (2x+1)^2 < 0.$$

Factorise as a difference of two squares with $a = x-3$, $b = 2x+1$:

$$\big[(x-3)-(2x+1)\big]\big[(x-3)+(2x+1)\big] < 0 \iff (-x-4)(3x-2) < 0.$$

Multiplying by -1 (reversing the inequality): $(x+4)(3x-2) > 0$. The roots are $x = -4$ and $x = \frac{2}{3}$. The product is positive outside the roots:

$$S = \left]-\infty\,;\,-4\right[\cup \left]\tfrac{2}{3}\,;\,+\infty\right[.$$

Check: $x = 0$: $|-3| = 3 > |1| = 1$, so $0 \notin S$ ✓; $x = 1$: $|-2| = 2 < |3| = 3$, so $1 \in S$ ✓.

WARNING **Never square when a side may be negative**

Squaring is valid only when both sides are known to be non-negative. From $-3 < 2$ one cannot deduce $9 < 4$! So for an inequality like $|x-1| < 2x+3$, you must first impose the condition $2x+3 \geq 0$ (a modulus is never less than a negative number), and only then square. Forgetting this condition introduces false solutions — a classic GCE trap.

5.5 The Critical Value (Case) Method

When moduli are added rather than compared, squaring does not help. We then use the definition of $|x|$ itself: each modulus changes its formula at the point where its contents vanish. These points are called **critical values**; they split the number line into cases.

METHOD **Critical value method**

1. Find the zero of each modulus: these are the critical values.
2. Arrange them in increasing order; they divide $\mathbb{R}$ into intervals (cases).
3. On each interval, replace every modulus by its correct expression ($+(\cdots)$ or $-(\cdots)$) using a sign table.
4. Solve the resulting ordinary inequality on that interval, and intersect the answer with

the interval.

5. Take the union of the partial solutions over all cases.

WORKED EXAMPLE 15 **Solving $|x-1|+|x+2| \le 4$**

Solve in $\mathbb{R}$: $|x-1|+|x+2| \le 4$.

SOLUTION

Solution. The critical values are $x = 1$ (from $|x-1|$) and $x = -2$ (from $|x+2|$). Sign table:

x	$-\infty$		-2		1	$+\infty$
$x-1$		$-$		$-$	0	$+$
$x+2$		$-$	0	$+$		$+$

Case 1: $x < -2$. Then $|x-1| = -(x-1)$ and $|x+2| = -(x+2)$:

$$-(x-1)-(x+2) \le 4 \iff -2x-1 \le 4 \iff x \ge -\tfrac{5}{2}.$$

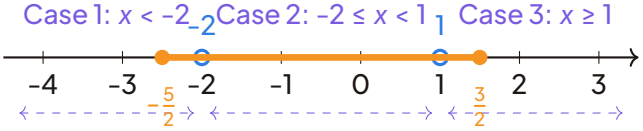
Intersecting with $x < -2$: $\left[-\tfrac{5}{2}; -2\right[$.
Case 2: $-2 \le x < 1$. Then $|x-1| = -(x-1)$, $|x+2| = x+2$:

$$-(x-1)+(x+2) \le 4 \iff 3 \le 4,$$

which is always true. The whole interval $[-2; 1[$ is a solution.
Case 3: $x \ge 1$. Then $|x-1| = x-1$, $|x+2| = x+2$:

$$(x-1)+(x+2) \le 4 \iff 2x+1 \le 4 \iff x \le \tfrac{3}{2}.$$

Intersecting with $x \ge 1$: $\left[1; \tfrac{3}{2}\right]$.
Union: $S = \left[-\tfrac{5}{2}; -2\right[\cup [-2; 1[\cup \left[1; \tfrac{3}{2}\right] = \left[-\tfrac{5}{2}; \tfrac{3}{2}\right]$.
Check: $x = 0$: $|-1|+|2| = 3 \le 4$ ✓; $x = 2$: $1+4 = 5 \not\le 4$ ✓; endpoint $x = -\tfrac{5}{2}$: $\tfrac{7}{2}+\tfrac{1}{2} = 4 \le 4$ ✓ (included, as the inequality is non-strict).



The critical values -2 and 1 split the line into three cases; the union of the partial solutions is the shaded interval $\left[-\tfrac{5}{2}; \tfrac{3}{2}\right]$.

WARNING **Intersect before you unite**

In each case, the solution of the simplified inequality is only valid on that case's interval. A frequent error is to keep $x \geq -\frac{5}{2}$ from Case 1 without intersecting with $x < -2$. Always intersect within the case, then take the union across cases.

5.6 Modulus Combined with Rational Expressions

GCE questions often mix a modulus with a fraction, e.g. $\left|\frac{x-2}{x+1}\right| < 2$. Two safe strategies exist.

Strategy A (quotient of moduli). Use $\left|\frac{A}{B}\right| = \frac{|A|}{|B|}$ with $B \neq 0$:

$$\left|\frac{x-2}{x+1}\right| < 2 \iff \frac{|x-2|}{|x+1|} < 2 \iff |x-2| < 2|x+1| \quad (x \neq -1).$$

Both sides are non-negative, so we square: $(x-2)^2 < 4(x+1)^2$, i.e. $(x-2)^2 - 4(x+1)^2 < 0$, which factorises as $[(x-2)-2(x+1)][(x-2)+2(x+1)] = (-x-4)(3x) < 0$, or $x(x+4) > 0$. Hence $x < -4$ or $x > 0$, and $x = -1$ is already excluded. So $S =]-\infty; -4[\cup]0; +\infty[$.

Strategy B (sign tables per case). If the inequality has the form $\frac{|A|}{B} < c$ with B not in a modulus, split at the zero of A and at the zero of B, remove the modulus per case, and solve each rational inequality with a sign table, remembering the excluded value.

EXEMPLE **Verification as an exam technique**

Whatever method you use, finish by testing sample values: one inside your proposed solution set, one outside, and each endpoint. This takes less than a minute and catches sign slips, forgotten intersections and wrongly included endpoints. In Strategy A above: $x = 1$ gives $\frac{1}{2} < 2$ ✓ (in S); $x = -2$ gives $4 \not< 2$ ✓ (not in S); $x = -4$ gives $2 \not< 2$ ✓ (endpoint correctly excluded).

KEY POINTS **Key points of the section**

- $|A| < b \iff -b < A < b$; $\quad |A| > b \iff A < -b$ or $A > b$ (a union).
- $|A| < |B| \iff A^2 < B^2$: squaring is legal because both sides are ≥ 0.
- Never square an inequality when one side may be negative; impose the sign condition first.
- For sums of moduli, use the critical value method: zeros of each modulus → cases → solve per case → intersect, then unite.
- Always present the answer in interval notation and verify with test values.

EXERCISE 6 **Practice**

Solve in $\mathbb{R}$ and give each answer in interval notation:

1. $|4x - 5| \le 7$;
2. $|2x + 3| > 1$;
3. $|x + 1| \ge |3x - 5|$;
4. $|x| + |x - 2| < 6$;
5. $\left|\dfrac{2x - 1}{x + 2}\right| \le 3$.

SOLUTION **Exercises 1–5**

1. $-7 \le 4x - 5 \le 7 \iff -2 \le 4x \le 12 \iff -\frac{1}{2} \le x \le 3$; $S = \left[-\frac{1}{2}; 3\right]$.
2. $2x + 3 < -1$ or $2x + 3 > 1 \iff x < -2$ or $x > -1$; $S =]-\infty; -2[\cup]-1; +\infty[$.
3. Square: $(x + 1)^2 \ge (3x - 5)^2 \iff (x + 1)^2 - (3x - 5)^2 \ge 0 \iff (-2x + 6)(4x - 4) \ge 0 \iff (x - 3)(x - 1) \le 0$; $S = [1; 3]$.
4. Critical values 0 and 2. Case $x < 0$: $-x - (x - 2) < 6 \iff x > -2$, giving $]-2; 0[$. Case $0 \le x < 2$: $x - (x - 2) = 2 < 6$ always true, giving $[0; 2[$. Case $x \ge 2$: $2x - 2 < 6 \iff x < 4$, giving $[2; 4[$. Union: $S =]-2; 4[$.
5. $|2x - 1| \le 3|x + 2|$, $x \ne -2$. Square: $(2x - 1)^2 \le 9(x + 2)^2 \iff (2x - 1)^2 - 9(x + 2)^2 \le 0 \iff (-x - 7)(5x + 5) \le 0 \iff (x + 7)(x + 1) \ge 0$; hence $x \le -7$ or $x \ge -1$ (-2 not concerned). $S =]-\infty; -7] \cup [-1; +\infty[$.

With these algebraic techniques mastered, we are ready for the final section, where the same inequalities are solved graphically and where all the skills of the chapter come together in complete curve sketching.

6 GRAPHICAL SOLUTIONS OF INEQUALITIES AND COMPLETE CURVE SKETCHING

In the previous section we solved absolute-value inequalities algebraically, by squaring, splitting into cases and using the double-inequality trick. Algebra is powerful, but it has a price: heavy computation, and no picture of what is really happening. In this final section we take the opposite route. We learn to read solutions directly from a graph, and then we assemble everything studied in this chapter — domain, limits, asymptotes, variation, concavity, symmetry — into one complete, examinable skill: the **full study and sketching of a function**, the flagship question of GCE Further Mathematics Paper 2.

6.1 The Graphical Meaning of an Inequality

EXPERIMENT **One picture is worth a page of algebra**

Draw on the same axes the parabola $y = x^2$ and the line $y = x + 2$. They meet where $x^2 = x + 2$, i.e. at $x = -1$ and $x = 2$. Now ask: where is the parabola above the line? Looking at the picture, the parabola dominates outside the interval $[-1, 2]$. You have just solved $x^2 > x + 2$ graphically: $x < -1$ or $x > 2$. No sign table was needed — only the two intersection abscissae and a glance at the picture.

PROPERTY **Graphical meaning of $f(x) > g(x)$**

Let f and g be two functions with curves C_f and C_g drawn on the same axes.

- $f(x) = g(x)$ at the **abscissae of the intersection points** of the two curves;
- $f(x) > g(x)$ exactly on the set of x-values where C_f lies **strictly above** C_g;
- $f(x) < g(x)$ exactly where C_f lies strictly below C_g.

The solution set is always a set of values of x: it is read on the x-axis, never on the y-axis.

METHOD **Solving $f(x) > g(x)$ graphically**

1. Draw (or use the given) curves C_f and C_g on the same axes.
2. Find the abscissae of the intersection points, either by reading them off the graph or by solving $f(x) = g(x)$ algebraically.
3. On each interval determined by these abscissae, decide which curve is on top.
4. **Project** the relevant portions of the curves onto the x-axis and write the solution set as a union of intervals.

WARNING **Project onto the x-axis!**

The most frequent error in graphical questions is to read the answer on the y-axis. An inequality in x has a solution set made of values of x. After locating the intersection points, drop vertical lines down to the x-axis and read the intervals there. Writing "$y > 2$" as a final answer to "solve $f(x) > 2$" costs marks at the GCE.

6.2 Graphical Solution of Absolute-Value Inequalities

6.3 The curve $y = |f(x)|$

To solve $|f(x)| < k$ graphically we first need the graph of $y = |f(x)|$. Recall that

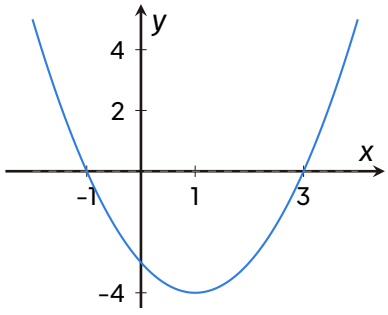
$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0, \\ -f(x) & \text{if } f(x) < 0. \end{cases}$$

So wherever f is non-negative, the two curves coincide; wherever f is negative, taking the modulus changes the sign of the ordinate, which geometrically is a **reflection in the x-axis**.

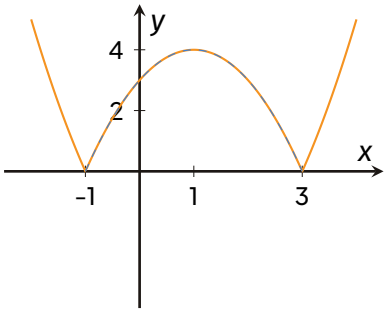
METHOD **Sketching $y = |f(x)|$ from $y = f(x)$**

1. Sketch $y = f(x)$ and mark its x-intercepts.
2. Keep unchanged every part of the curve lying on or above the x-axis.
3. Reflect in the x-axis every part lying below the x-axis (it flips up).
4. The resulting curve never goes below the x-axis; at the intercepts it typically forms a sharp "corner" (a cusp-like point where the curve is not differentiable).

$y = f(x) = x^2 - 2x - 3$



$y = |f(x)|$



Left: the parabola $y = f(x)$. Right: $y = |f(x)|$; the dashed arc shows the original negative part, which has been reflected upwards in the x-axis.

6.4 Solving $|f(x)| < k$ and $|f(x)| > k$ graphically

PROPERTY **Modulus equations and horizontal lines**

Let $k > 0$. The solutions of $|f(x)| = k$ are exactly the abscissae of the intersection points of the curve $y = f(x)$ (or equivalently $y = |f(x)|$) with the horizontal lines $y = k$ and $y = -k$.

Consequently:

- $|f(x)| < k$ holds where the curve $y = |f(x)|$ lies below the line $y = k$;
- $|f(x)| > k$ holds where the curve $y = |f(x)|$ lies above the line $y = k$.

(The proof is immediate from $|f(x)| < k \iff -k < f(x) < k$, seen in the previous section; it is not examinable but must be understood.)

WORKED EXAMPLE 16 **Solving $|x^2 - 2x - 3| < 2$ graphically**

Using the graph of $y = |x^2 - 2x - 3|$, solve the inequality $|x^2 - 2x - 3| < 2$.

SOLUTION

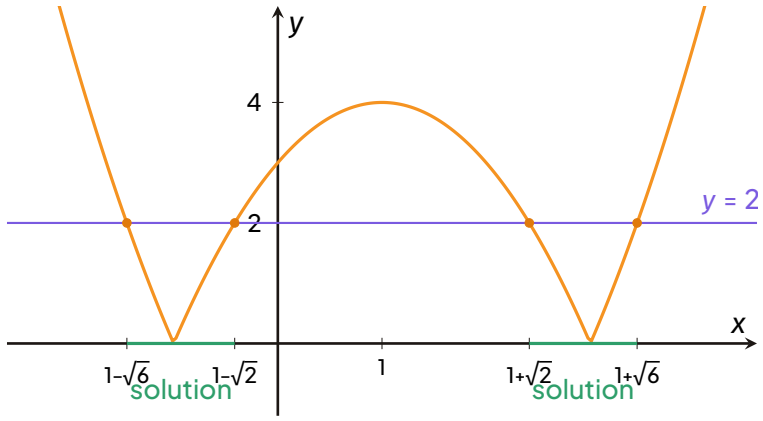
Step 1: find the intersections with $y = 2$. We solve $|x^2 - 2x - 3| = 2$, i.e.

$$x^2 - 2x - 3 = 2 \quad \text{or} \quad x^2 - 2x - 3 = -2.$$

The first gives $x^2 - 2x - 5 = 0$, so $x = 1 \pm \sqrt{6}$. The second gives $x^2 - 2x - 1 = 0$, so $x = 1 \pm \sqrt{2}$.
Numerically: $1 - \sqrt{6} \approx -1.45$, $1 - \sqrt{2} \approx -0.41$, $1 + \sqrt{2} \approx 2.41$, $1 + \sqrt{6} \approx 3.45$.
Step 2: read the graph. The modulus curve lies below the line $y = 2$ between the first and second intersections, and again between the third and fourth.
Step 3: project onto the x-axis.

$$S = \left(1 - \sqrt{6},\ 1 - \sqrt{2}\right) \cup \left(1 + \sqrt{2},\ 1 + \sqrt{6}\right).$$

This agrees with the algebraic solution obtained by the double-inequality method — a good cross-check.



Graphical solution of $|x^2 - 2x - 3| < 2$: the curve $y = |f(x)|$ is below $y = 2$ exactly above the thick green segments of the x-axis.

6.5 Comparing two modulus curves: $|f(x)| < |g(x)|$

The same principle applies with two curves: sketch $y = |f(x)|$ and $y = |g(x)|$ on the same axes, find the abscissae of their intersection points (solve $|f(x)| = |g(x)|$, i.e. $f(x) = g(x)$ or $f(x) = -g(x)$), then read where the first modulus curve is below the second.

EXEMPLE **A quick comparison**

Solve graphically $|x - 1| < |x + 1|$. The two V-shaped curves meet where $x - 1 = -(x + 1)$, i.e. $x = 0$ (the case $x - 1 = x + 1$ is impossible). For $x > 0$ the line $y = |x - 1|$ lies below $y = |x + 1|$. Hence $S = (0, +\infty)$. Geometrically this says: the points closer to 1 than to -1 are exactly the points of the right half-line — a fact you also met with perpendicular bisectors in pure geometry.

6.6 The curve $y = f(|x|)$ (a useful GCE extra)

PROPERTY **The graph of $y = f(|x|)$**

For $x \geq 0$, $f(|x|) = f(x)$, so the graph coincides with the right half of C_f. For $x < 0$, $f(|x|) = f(-x)$, which is the mirror image of the right half in the y-axis. Hence $y = f(|x|)$ is always an **even** function: keep the part of C_f with $x \geq 0$, discard the left half, and reflect the kept part in the y-axis.

WARNING **Do not confuse $|f(x)|$ with $f(|x|)$**

These are two different transformations. $y = |f(x)|$ reflects the negative ordinates in the x-axis (vertical flip). $y = f(|x|)$ reflects the right half-graph in the y-axis (horizontal flip) and produces an even curve. A frequent mistake is to reflect the left half of C_f instead of the right half: remember that $f(|x|)$ at $x = -3$ uses the value of f at $+3$, so it is the right half that is copied.

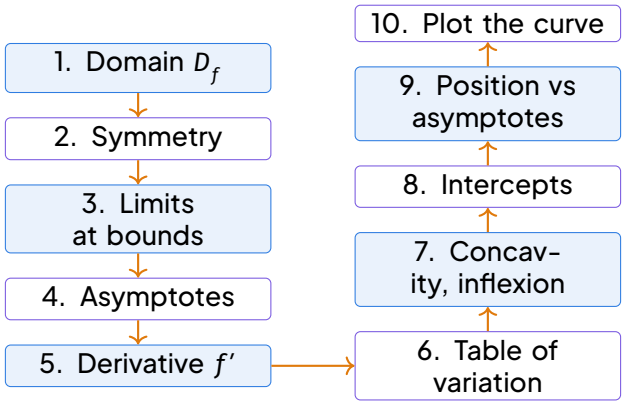
6.7 The Complete Curve-Sketching Algorithm

Every lesson of this chapter now comes together. A full study of a function, as demanded in Paper 2, follows a fixed checklist. Mastering the order of the steps is what separates a confident candidate from a lost one.

METHOD **The ten-step complete study of a function**

1. Domain: find the domain of definition D_f (denominators, square roots, logarithms).

2. Symmetry: test $f(-x)$; if f is even or odd, study on half the domain and deduce the rest (axis or centre of symmetry). Also look for other centres/axes via $f(2a - x) = f(x)$ or $f(a + t) + f(a - t) = 2b$.
3. Limits at the bounds of D_f, including $\pm\infty$.
4. Asymptotes: vertical (infinite limits), horizontal (finite limits at infinity), oblique $y = ax + b$ with $a = \lim f(x)/x$, $b = \lim[f(x) - ax]$.
5. Derivative: compute f', solve $f'(x) = 0$, find stationary points.
6. Table of variation: sign of f', arrows, remarkable values.
7. Concavity: sign of f''; points of inflexion where f'' changes sign.
8. Intercepts: y-intercept $f(0)$; x-intercepts from $f(x) = 0$ (exact or approximate).
9. Position of the curve relative to its asymptotes (sign of $f(x) - (ax + b)$) and infinite branches.
10. Plot: draw asymptotes first, then remarkable points, then the curve respecting variation, concavity and position.



The ten-step curve-sketching algorithm: follow the arrows and no GCE question will surprise you.

6.8 Full Worked Study: a Rational Function with an Oblique Asymptote

WORKED EXAMPLE 17 **Complete study of** $f(x) = \dfrac{x^2 - 3x + 3}{x - 2}$

Let $f(x) = \dfrac{x^2 - 3x + 3}{x - 2}$ and let C be its curve.

1. Study f completely (domain, limits, asymptotes, variation, position).
2. Sketch C.
3. Deduce the number of solutions of $f(x) = m$ according to the values of the real number m.
4. Solve graphically $|f(x)| < 1$.

SOLUTION

1. Domain. $D_f = \mathbb{R} \setminus \{2\}$. No symmetry (the domain is not symmetric about 0).
2. Limits. $\lim\limits_{x \to 2^-} f(x) = -\infty$ and $\lim_{x \to 2^+} f(x) = +\infty$ (numerator at $x = 2$ equals $1 > 0$). At infinity, $f(x) \sim x$, so $\lim_{x \to \pm\infty} f(x) = \pm\infty$.
3. Asymptotes. The line $x = 2$ is a vertical asymptote. Polynomial division gives

$$f(x) = x - 1 + \frac{1}{x-2},$$

so $\lim_{x \to \pm\infty}[f(x) - (x-1)] = 0$: the line $y = x - 1$ is an **oblique asymptote**. Moreover $f(x) - (x-1) = \frac{1}{x-2}$: the curve is above the asymptote for $x > 2$ and below it for $x < 2$.
4. Derivative and variation.

$$f'(x) = 1 - \frac{1}{(x-2)^2} = \frac{(x-2)^2 - 1}{(x-2)^2} = \frac{(x-1)(x-3)}{(x-2)^2}.$$

$f'(x) = 0$ at $x = 1$ and $x = 3$; $f(1) = -1$ (local maximum) and $f(3) = 3$ (local minimum). Note the amusing fact $f(3) > f(1)$: the two branches are separate.

x	$-\infty$		1		2		3		$+\infty$
$f'(x)$		$+$	0	$-$	‖	$-$	0	$+$	
$f(x)$	$-\infty$	↗	-1	↘	‖	↘	3	↗	$+\infty$

(On $(3, +\infty)$, $f' > 0$ and f rises from 3 to $+\infty$; on $(-\infty, 1)$, f rises from $-\infty$ to -1.)
5. Concavity. $f''(x) = \frac{2}{(x-2)^3}$: concave down on $(-\infty, 2)$, concave up on $(2, +\infty)$; no inflexion point since f'' never vanishes (the concavity changes across the asymptote, not at a point of the curve).
6. Intercepts. $f(0) = -\frac{3}{2}$. The numerator $x^2 - 3x + 3$ has discriminant $9 - 12 = -3 < 0$: no x-intercept.
7. Sketch. Draw the asymptotes $x = 2$ and $y = x - 1$, plot $(1, -1)$, $(3, 3)$, $(0, -1.5)$, then the two branches respecting the table and the position relative to the oblique asymptote.
8. Number of solutions of $f(x) = m$. Reading horizontal lines $y = m$ across the sketch:

- if $-1 < m < 3$: no solution;
- if $m = -1$ or $m = 3$: exactly one solution;
- if $m < -1$ or $m > 3$: exactly two solutions.

9. Solving $|f(x)| < 1$. Reflect the negative part of C in the x-axis to get $y = |f(x)|$, then cut with $y = 1$: $|f(x)| = 1 \iff f(x) = 1$ or $f(x) = -1$. From $f(x) = 1$: $x^2 - 3x + 3 = x - 2$, i.e. $x^2 - 4x + 5 = 0$, no real root. From $f(x) = -1$: $x^2 - 3x + 3 = -(x - 2)$, i.e. $x^2 - 2x + 1 = 0$, so $x = 1$ (double). The curve $y = |f(x)|$ only touches $y = 1$ at $x = 1$ and stays below it nowhere else… checking the variation: on $(-\infty, 2)$, $|f|$ ranges over $[1, +\infty)$ except near $x = 1$ where $|f(1)| = 1$; hence $|f(x)| < 1$ has no solution: $S = \varnothing$. The graph confirms it instantly.

WARNING **Two classic traps in full studies**

- Forgetting the position of the curve relative to the oblique asymptote: the sign of $f(x) - (ax + b)$ is a one-line computation and is explicitly demanded by many mark schemes.
- Concluding "no inflexion point" too quickly, or claiming one at $x = 2$ here: an inflexion point must be a point of the curve; f is not even defined at $x = 2$.

DID YOU KNOW? **Why Paper 2 loves this question**

A complete function study tests, in a single item, limits, differentiation, asymptotes, inequalities and graph reading — nearly the whole chapter. Cameroonian candidates who rehearse the ten-step checklist on three or four functions before the examination routinely turn this long question into their highest-scoring one. Treat the checklist as a ritual: same order, same table, same care with the axes, every time.

EXERCISE 7 **Practice**

1. Using the graph of $y = |x^2 - 2x - 3|$ drawn in this section, solve $|x^2 - 2x - 3| > 2$.
2. From the graph of $y = x^3 - 3x$, sketch $y = |x^3 - 3x|$ and $y = |x|^3 - 3|x|$, stating precisely which parts are kept or reflected.
3. Carry out the complete study of $g(x) = \dfrac{x^2 + x - 2}{x - 1}$ (oblique asymptote $y = x + 2$), then deduce the number of solutions of $g(x) = m$ and solve graphically $|g(x)| < 2$.

SOLUTION **Exercise 1**

1. The curve is above $y = 2$ outside the intervals found in the worked example: $S = (-\infty, 1 - \sqrt{6}) \cup (1 - \sqrt{2}, 1 + \sqrt{2}) \cup (1 + \sqrt{6}, +\infty)$.

2. For $y = |x^3 - 3x|$: keep the parts with $x \in [-\sqrt{3}, 0] \cup [\sqrt{3}, +\infty)$, reflect the rest in the x-axis. For $y = |x|^3 - 3|x| = f(|x|)$: keep the right half ($x \geq 0$) of the cubic and reflect it in the y-axis; the result is even, with minima at $x = \pm 1$.

3. $D_g = \mathbb{R} \setminus \{1\}$; $g(x) = x + 2$ exactly (since $x^2 + x - 2 = (x - 1)(x + 2)$)! The curve is the line $y = x + 2$ with the single point $(1, 3)$ removed — a "hole", not an asymptote. $g(x) = m$ has one solution for $m \neq 3$, none for $m = 3$. $|g(x)| < 2 \iff -2 < x + 2 < 2$ with $x \neq 1$: $S = (-4, 0)$.

KEY POINTS **Key points of the section**

- $f(x) > g(x)$ where C_f is above C_g; solutions are read on the x-axis.
- $y = |f(x)|$: keep the upper part, reflect the lower part in the x-axis. $y = f(|x|)$: keep the right half, reflect it in the y-axis (even function).
- $|f(x)| = k$ at the intersections with $y = k$ and $y = -k$; then read above/below.
- The ten-step checklist (domain, symmetry, limits, asymptotes, f', variation, f'', intercepts, position, plot) is the backbone of every Paper 2 curve-sketching question.
- A finished sketch is a tool: use it to count roots, solve inequalities and discuss positions.

7 INTEGRATION ACTIVITY

7.1 Aims of the activity

This integration activity closes the chapter by mobilising, in one authentic optimisation problem, every skill developed in Lessons 66 to 76. Working in teams, you will:

- determine the **domain of definition** of a rational-type function and compute its **limits at the bounds** of that domain;
- prove the existence of an **oblique asymptote** and study the **position of the curve** relative to it (infinite branches);
- compute a derivative, discuss **stationary points**, and build a complete **table of variation**;
- study **concavity** through the sign of the second derivative and decide whether a **point of inflexion** exists;
- solve an **absolute-value inequality** both algebraically and graphically, and interpret the solution set in context;
- produce a complete, well-annotated **curve sketch** and defend it before the class.

KEY POINTS **Skills being assessed**

Domain → limits → asymptotes → derivative and variation → concavity → sketch → inequality solving. This is exactly the chain of reasoning expected in the GCE Advanced Level Further Mathematics examination: a full function study is never a list of disconnected calculations, but a single coherent argument in which each step feeds the next.

7.2 Situation

Optimising a Water Tank's Profit Curve in Buea.

Madam Eposi runs a water-vending business near Molyko, Buea. Every day she fills her tank from the Cameroon Water Utilities network and resells water by the jerrycan to students of the University of Buea. After several months of record keeping, her elder brother, a Further Mathematics student, models her **daily profit** $P(x)$, measured in thousands of FCFA, as a function of the quantity x she sells, measured in hundreds of litres:

$$P(x) = \frac{-x^2 + 10x + 7}{x + 1}, \qquad x \geq 0.$$

The model reflects two realities of the business: moderate sales bring good margins, but beyond a certain level the extra costs (hiring an extra carrier, buying fuel for the pump during power cuts, discounts to attract hostels) eat into the profit, which eventually falls.

Madam Eposi wants answers to three practical questions:

- What quantity of water maximises the daily profit, and what is that maximum profit?

- For large sales, the profit decreases almost along a straight line: which line, and does the profit curve lie above or below it?
- For peace of mind, she accepts any sales level for which the profit stays within 2 thousand FCFA of the maximum. What is the corresponding range of daily sales?

Your team is hired as the consulting mathematicians. You must produce a poster containing the full study of the function P and a clear, justified answer to each question.

7.3 Tasks

1. **Domain and bounds.** Justify that the natural domain of P is $\mathbb{R} \setminus \{-1\}$ and state the domain of the model. Compute $P(0)$ and $\lim\limits_{x \to +\infty} P(x)$. Interpret each result for the business.

2. **Oblique asymptote and position.** Show that for all $x \geq 0$,

$$P(x) = -x + 11 - \frac{4}{x+1}.$$

   Deduce that the line $(\Delta) : y = -x + 11$ is an oblique asymptote to the curve (C) of P, and determine the position of (C) relative to (Δ).

3. **Variation and maximum.** Compute $P'(x)$ and show that $P'(x) = \dfrac{4}{(x+1)^2} - 1$. Solve $P'(x) = 0$ on $[0, +\infty[$, build the table of variation of P, and deduce the maximum profit and the quantity sold that achieves it.

4. **Concavity.** Compute $P''(x)$, study its sign on $[0, +\infty[$, state the concavity of (C) and decide whether (C) has a point of inflexion.

5. **Intercepts and sketch.** Find the intercepts of (C) with the axes (give surd form where necessary). Sketch (C), the asymptote (Δ), the maximum point and the intercepts on the same axes.

6. **The tolerance band.** Let $P_{\max}$ denote the maximum profit. Solve algebraically the inequality

$$\left|P(x) - P_{\max}\right| \leq 2 \qquad (x \geq 0),$$

   then verify your solution set graphically by drawing the horizontal lines $y = P_{\max} - 2$ and $y = P_{\max} + 2$ on your sketch. Interpret the solution set for Madam Eposi.

7. **Poster and critique.** Present your sketch and solution set on a poster. Each team checks another team's poster against the marking scheme: correct domain, correct asymptote with position, correct table of variation, correct concavity, correct solution set.

7.4 Model answer

1. Domain and bounds.

P is a rational function (quotient of two polynomials); it is defined unless its denominator vanishes, i.e. unless $x + 1 = 0$, that is $x = -1$. The **natural domain** is therefore $\mathbb{R} \setminus \{-1\}$. Since a quantity of water sold cannot be negative, the domain of the model is $D = [0, +\infty[$. Note that the forbidden value $x = -1$ lies outside the model's domain, so no vertical asymptote is relevant to the business study.

At the left bound of D:

$$P(0) = \frac{-0 + 0 + 7}{0 + 1} = 7.$$

At the right bound, factorise the highest powers of x (standard technique for limits of rational functions at infinity): for $x > 0$,

$$P(x) = \frac{x^2\left(-1 + \frac{10}{x} + \frac{7}{x^2}\right)}{x\left(1 + \frac{1}{x}\right)} = x \cdot \frac{-1 + \frac{10}{x} + \frac{7}{x^2}}{1 + \frac{1}{x}}.$$

As $x \to +\infty$, the bracketed fraction tends to $\frac{-1}{1} = -1$, while the factor x tends to $+\infty$. Hence

$$\lim_{x \to +\infty} P(x) = -\infty.$$

Interpretation: with no sales, a fixed income (for instance renting out empty cans) yields 7 thousand FCFA, i.e. 7000 FCFA; excessively large sales drive the profit down without bound because of the extra costs. The curve therefore has an **infinite branch** at $+\infty$, whose nature is determined in Task 2.

2. Oblique asymptote and position.

We perform the polynomial division of $-x^2 + 10x + 7$ by $x + 1$. Since

$$(x + 1)(-x + 11) = -x^2 + 11x - x + 11 = -x^2 + 10x + 11,$$

we obtain

$$-x^2 + 10x + 7 = (x + 1)(-x + 11) - 4.$$

Dividing both sides by $x + 1$ (valid for $x \neq -1$, in particular for all $x \geq 0$):

$$P(x) = -x + 11 - \frac{4}{x + 1}.$$

Now examine the gap between the curve and the line (Δ) : $y = -x + 11$:

$$P(x) - (-x + 11) = -\frac{4}{x + 1} \xrightarrow[x \to +\infty]{} 0.$$

By definition, (Δ) is an **oblique asymptote** of (C) at $+\infty$: this is the "straight line" Madam Eposi observed.

For the **position of the curve** relative to the asymptote, study the sign of the gap: for all $x \geq 0$, $x + 1 > 0$, so

$$P(x) - (-x + 11) = -\frac{4}{x + 1} < 0.$$

Therefore (C) lies strictly **below** (Δ) on the whole domain: the profit approaches the line from underneath, never reaching it.

3. Variation and maximum.

Differentiate the divided form $P(x) = -x + 11 - 4(x + 1)^{-1}$ term by term:

$$P'(x) = -1 - 4 \cdot \left(-(x + 1)^{-2}\right) = -1 + \frac{4}{(x + 1)^2} = \frac{4 - (x + 1)^2}{(x + 1)^2}.$$

Factorising the numerator as a difference of two squares, $4-(x+1)^2=\big(2-(x+1)\big)\big(2+(x+1)\big)$:

$$P'(x)=\frac{(1-x)(x+3)}{(x+1)^2}.$$

On $[0,+\infty[$, $(x+1)^2>0$ and $x+3>0$, so $P'(x)$ has the sign of $1-x$. The equation $P'(x)=0$ reduces to $1-x=0$, i.e. $x=1$ (the root $x=-3$ is outside the domain). Thus $P'(x)>0$ on $[0,1[$, $P'(1)=0$, and $P'(x)<0$ on $]1,+\infty[$. The only **stationary point** is at $x=1$, and

$$P(1)=-1+11-\frac{4}{2}=8.$$

x	0		1		$+\infty$
$P'(x)$		$+$	0	$-$	
$P(x)$	7	$\nearrow$	8	$\searrow$	$-\infty$

Since P' changes sign from $+$ to $-$ at $x=1$, the stationary point is a **maximum**. Interpretation: the maximum daily profit is $P_{\max}=8$ thousand FCFA, i.e. 8000 FCFA, obtained by selling $x=1$ hundred litres, that is 100 litres per day.

4. Concavity.

Differentiate $P'(x)=-1+4(x+1)^{-2}$:

$$P''(x)=4\cdot\big(-2(x+1)^{-3}\big)=-\frac{8}{(x+1)^3}.$$

For all $x\geq 0$, $(x+1)^3>0$, so $P''(x)<0$: the curve is **concave down** on the whole domain (its tangents all lie above it). Since P'' never vanishes and never changes sign on $[0,+\infty[$, (C) has **no point of inflexion**. This is consistent with Task 3: a function that is concave down around a stationary point has a maximum there.

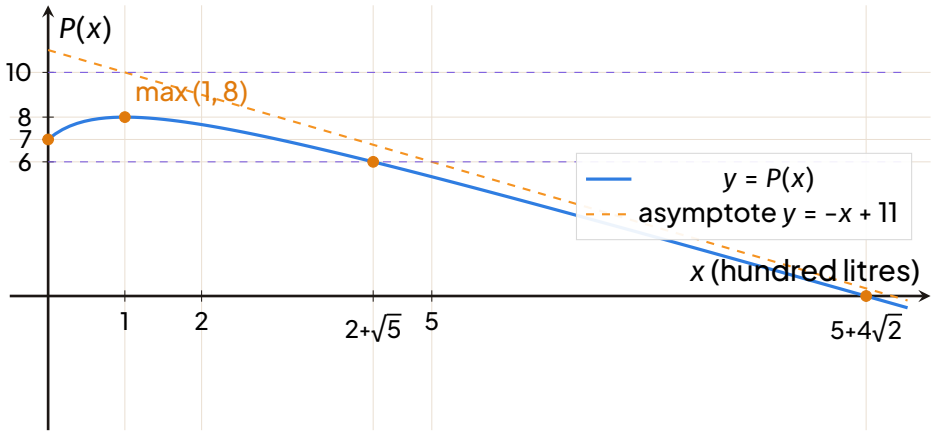
5. Intercepts and sketch.

y-intercept: $P(0)=7$, giving the point $(0,7)$.

x-intercepts: solve $P(x)=0$, i.e. $-x^2+10x+7=0$, equivalently $x^2-10x-7=0$. The discriminant is $\Delta=100+28=128=(8\sqrt{2})^2$, so

$$x=\frac{10\pm 8\sqrt{2}}{2}=5\pm 4\sqrt{2}.$$

Only $x=5+4\sqrt{2}\approx 10.66$ lies in the domain $[0,+\infty[$ (the other root $5-4\sqrt{2}\approx -0.66$ is rejected). Interpretation: the profit becomes negative beyond about 1066 litres sold per day.



The profit curve, its oblique asymptote, and the tolerance band $6 \le y \le 10$.

6. The tolerance band.

Algebraic solution. With $P_{max} = 8$, the key equivalence for absolute values, $|u| \le a \iff -a \le u \le a$, gives

$$|P(x) - 8| \le 2 \iff -2 \le P(x) - 8 \le 2 \iff 6 \le P(x) \le 10.$$

The table of variation shows $P(x) \le 8 < 10$ for every $x \ge 0$, so the right-hand inequality $P(x) \le 10$ is automatically satisfied. Only $P(x) \ge 6$ remains:

$$-x + 11 - \frac{4}{x+1} \ge 6 \iff 5 - x \ge \frac{4}{x+1}.$$

Since $x \ge 0$, we have $x + 1 > 0$, so multiplying both sides by $x + 1$ preserves the inequality:

$$(5 - x)(x + 1) \ge 4 \iff -x^2 + 4x + 5 \ge 4 \iff x^2 - 4x - 1 \le 0.$$

The quadratic $x^2 - 4x - 1$ has discriminant $16 + 4 = 20$ and roots

$$x = \frac{4 \pm \sqrt{20}}{2} = 2 \pm \sqrt{5}.$$

A quadratic with positive leading coefficient is negative between its roots, so

$$x \in [\,2 - \sqrt{5},\ 2 + \sqrt{5}\,].$$

Finally, intersect with the domain $x \ge 0$; since $2 - \sqrt{5} \approx -0.24 < 0$:

$$\boxed{S = [\,0,\ 2 + \sqrt{5}\,] \approx [\,0,\ 4.24\,]}$$

Graphical verification. The horizontal lines $y = P_{max} - 2 = 6$ and $y = P_{max} + 2 = 10$ delimit a band. On the sketch, the curve enters the band at $x = 0$ (since $P(0) = 7$) and stays inside it until it crosses the line $y = 6$ at $x = 2 + \sqrt{5} \approx 4.24$; beyond that it leaves the band from below. The graphical reading matches the algebraic solution set exactly.

Interpretation: Madam Eposi's profit stays within 2000 FCFA of her maximum as long as daily sales do not exceed about 424 litres; beyond that level, the profit drops more than 2000 FCFA below the best possible value.

WARNING **Common errors observed during the critique session**

- Forgetting to intersect the algebraic solution $[2 - \sqrt{5},\ 2 + \sqrt{5}]$ with the domain $x \geq 0$ of the model.
- Claiming an inflexion point "because there is a maximum": a maximum comes from P' changing sign, an inflexion point from P'' changing sign. Here $P''(x) < 0$ throughout the domain.
- Writing the asymptote as $y = -x + 11$ "by looking at the leading terms" without computing $P(x) - (-x + 11)$ and showing it tends to 0: this justification is examinable and carries marks.
- Multiplying an inequality by $x + 1$ without first noting that $x + 1 > 0$ on the domain, so the direction of the inequality is preserved.
- Solving $|P(x) - 8| \leq 2$ as only $P(x) \geq 6$ without first checking that $P(x) \leq 10$ holds automatically.

DID YOU KNOW? **Why a tolerance band?**

In quality control and business management, decisions are rarely based on an exact optimum: one accepts a tolerance interval around it. Writing "within a units of the target m" is precisely the inequality $|x - m| \leq a$, which is why absolute-value inequalities appear in economics, engineering and the GCE examination alike.

8 METHODS AND WORKED EXAMPLES

9 FURTHER CURVE SKETCHING AND INEQUALITIES

9.1 Domain of Definition and Limits at the Bounds

DEFINITION **Domain of Definition**

The **domain of definition** D_f of a function f is the set of all real numbers x for which the expression $f(x)$ makes sense (i.e. can be computed without violating the rules of real arithmetic).

PROPERTY **Common Restrictions on D_f**

For real-valued functions of a real variable:

- Denominator: $q(x) \neq 0$ for $f(x) = \frac{p(x)}{q(x)}$.
- Even root: $u(x) \geq 0$ for $f(x) = \sqrt[n]{u(x)}$ with n even.
- Logarithm: $u(x) > 0$ for $f(x) = \ln u(x)$ or $\log_a u(x)$.
- Tangent: $u(x) \neq \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$ for $f(x) = \tan u(x)$.
- Arcsine/Arccosine: $-1 \leq u(x) \leq 1$ for $f(x) = \arcsin u(x)$ or $\arccos u(x)$.

D_f is the intersection of all individual conditions, expressed as a union of intervals.

METHOD **Finding the Domain and Limits at its Bounds**

1. Identify the constraints. Ask: what could make $f(x)$ impossible to compute? (See the list above.)
2. Solve each constraint as an equation or inequality in x, then intersect all the conditions. Write D_f as a union of intervals.
3. Compute the limits at every bound of D_f:
   - At $\pm\infty$: compare degrees for rational functions; factorise the dominant term.
   - At each excluded finite value a: compute $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ separately

(sign of the denominator on each side).

4. Interpret: an infinite limit at a finite bound $x = a$ announces a **vertical asymptote**; a finite limit at $\pm\infty$ announces a **horizontal asymptote**.

Reflex: never write $\lim_{x\to a}$ when a is excluded from the domain — always use one-sided limits.

EXEMPLE **Domain and limits of a rational function with a radical**

Let $f(x) = \dfrac{\sqrt{x+1}}{x-2}$.

1. Determine the domain of definition D_f of f.
2. Compute the limits of f at the bounds of D_f and interpret the results graphically.

1. Domain of definition.
Two constraints act simultaneously:

- The square root requires $x + 1 \geq 0$, i.e. $x \geq -1$.
- The denominator requires $x - 2 \neq 0$, i.e. $x \neq 2$.

Both conditions must hold, so

$$D_f = [-1, 2) \cup (2, +\infty).$$

2. Limits at the bounds.
At $x = -1$ (included bound): we simply evaluate:

$$\lim_{x\to -1^+} f(x) = \frac{\sqrt{0}}{-1-2} = 0.$$

The curve starts at the point $(-1, 0)$.
At $x = 2$ (excluded bound): the numerator tends to $\sqrt{3} > 0$; the denominator tends to 0 with a sign depending on the side:

- If $x \to 2^-$, then $x - 2 < 0$, so $f(x) \to \dfrac{\sqrt{3}}{0^-} = -\infty$.
- If $x \to 2^+$, then $x - 2 > 0$, so $f(x) \to \dfrac{\sqrt{3}}{0^+} = +\infty$.

Interpretation: the line $x = 2$ is a **vertical asymptote**.
At $+\infty$: for $x > 0$, write $\sqrt{x+1} = \sqrt{x}\sqrt{1+\frac{1}{x}}$, so

$$f(x) = \frac{\sqrt{x}\sqrt{1+\frac{1}{x}}}{x\left(1-\frac{2}{x}\right)} = \frac{1}{\sqrt{x}} \cdot \frac{\sqrt{1+\frac{1}{x}}}{1-\frac{2}{x}} \xrightarrow[x\to+\infty]{} 0 \times \frac{1}{1} = 0.$$

Interpretation: the line $y = 0$ (the x-axis) is a **horizontal asymptote** at $+\infty$.

KEY POINTS **Key Points: Domain and Limits**

- D_f is an intersection of conditions; write it as a union of intervals.
- At a finite bound $a \notin D_f$, always compute one-sided limits $\lim_{x\to a^-}$ and $\lim_{x\to a^+}$.
- $\lim_{x\to a^\pm} f(x) = \pm\infty \implies$ vertical asymptote $x = a$.
- $\lim_{x\to\pm\infty} f(x) = b \in \mathbb{R} \implies$ horizontal asymptote $y = b$.

9.2 Asymptotes, Intercepts and Infinite Branches

PROPERTY **Types of Asymptotes**

Let (C) be the curve of f.

- Vertical asymptote $x = a$: $\lim_{x\to a^\pm} f(x) = \pm\infty$ (or $\mp\infty$).
- Horizontal asymptote $y = b$: $\lim_{x\to\pm\infty} f(x) = b$ (finite).
- Oblique asymptote $y = ax + b$ $(a \neq 0)$: $\lim_{x\to\pm\infty} \left[f(x) - (ax + b)\right] = 0$.

For a rational function $f(x) = \frac{P(x)}{Q(x)}$:

- $\deg P < \deg Q \implies y = 0$ horizontal asymptote.
- $\deg P = \deg Q \implies y = \frac{\text{leading coeff of } P}{\text{leading coeff of } Q}$ horizontal asymptote.
- $\deg P = \deg Q + 1 \implies$ oblique asymptote found by polynomial long division.
- $\deg P \geq \deg Q + 2 \implies$ no oblique asymptote; **parabolic branch** (study $\lim \frac{f(x)}{x}$).

METHOD **Asymptotes, Intercepts and Position Relative to Asymptotes**

1. Vertical asymptotes: at each excluded value $x = a$, if $\lim_{x\to a^\pm} f(x) = \pm\infty$, then $x = a$ is a vertical asymptote.
2. Horizontal asymptotes: if $\lim_{x\to\pm\infty} f(x) = b$ (finite), then $y = b$ is a horizontal asymptote at that end.
3. Oblique asymptotes (rational function with deg(num) = deg(den) + 1): perform polynomial long division to write $f(x) = ax + b + \dfrac{r(x)}{q(x)}$ with $\deg r < \deg q$. Since $\dfrac{r(x)}{q(x)} \to 0$, the line $y = ax + b$ is an oblique asymptote.
4. Position of the curve relative to the line: study the sign of the remainder $f(x) - (ax + b) =$

$\frac{r(x)}{q(x)}$. Positive ⇒ curve above; negative ⇒ curve below.

5. Parabolic branches: if $\lim_{x\to\pm\infty} f(x) = \pm\infty$ and no oblique asymptote, compute $\lim_{x\to\pm\infty} \frac{f(x)}{x}$:
   - If 0: parabolic branch in the direction of (Ox).
   - If $\pm\infty$: parabolic branch in the direction of (Oy).
6. Intercepts: y-intercept = $f(0)$ (if $0 \in D_f$); x-intercepts: solve $f(x) = 0$ (numerator = 0 for a rational function, provided the denominator is non-zero).

EXEMPLE **Oblique asymptote and relative position**

Let $f(x) = \frac{x^2 - 3x + 5}{x - 1}$ and let (C) be its curve.

1. Show that the line $(\Delta) : y = x - 2$ is an asymptote to (C) at $+\infty$ and at $-\infty$.
2. Study the position of (C) relative to (Δ).
3. Give all asymptotes and the intercepts of (C).

1. Oblique asymptote. Divide $x^2 - 3x + 5$ by $x - 1$:

$$x^2 - 3x + 5 = (x - 1)(x - 2) + 3.$$

(Check: $(x - 1)(x - 2) = x^2 - 3x + 2$, and $x^2 - 3x + 2 + 3 = x^2 - 3x + 5$. ✓) Hence

$$f(x) = x - 2 + \frac{3}{x - 1}.$$

Since $\lim_{x\to\pm\infty} \frac{3}{x - 1} = 0$, we get $\lim_{x\to\pm\infty} \left[f(x) - (x - 2)\right] = 0$: the line $y = x - 2$ is an **oblique asymptote** at both $+\infty$ and $-\infty$.

2. Relative position. The sign of the gap is the sign of

$$f(x) - (x - 2) = \frac{3}{x - 1}.$$

- If $x > 1$: $\frac{3}{x - 1} > 0$, so (C) is above (Δ).
- If $x < 1$: $\frac{3}{x - 1} < 0$, so (C) is below (Δ).

3. All asymptotes and intercepts.

$D_f = \mathbb{R} \setminus \{1\}$. At $x = 1$: the numerator equals $1 - 3 + 5 = 3 > 0$, and

$$\lim_{x\to 1^-} f(x) = -\infty, \qquad \lim_{x\to 1^+} f(x) = +\infty,$$

so $x = 1$ is a **vertical asymptote**.
y-intercept: $f(0) = \frac{5}{-1} = -5$; the curve passes through $(0, -5)$.
x-intercepts: solve $x^2 - 3x + 5 = 0$; $\Delta = 9 - 20 = -11 < 0$, so there is no x-intercept (the curve never crosses the x-axis; indeed $x^2 - 3x + 5 > 0$ for all x).

WARNING **Common Mistake: Oblique Asymptote**

Do not confuse the quotient of the division with the asymptote equation. If $f(x) = \frac{P(x)}{Q(x)}$ and $P(x) = Q(x)(ax + b) + R(x)$, then the oblique asymptote is $y = ax + b$, not $y = \frac{P(x)}{Q(x)}$. Always verify that $\deg R < \deg Q$.

9.3 Stationary Points and Variation of a Function

DEFINITION **Stationary (Critical) Point**

A point $x_0 \in D_f$ is a **stationary point** (or **critical point**) of f if $f'(x_0) = 0$ or $f'(x_0)$ does not exist (while f is continuous at x_0).

PROPERTY **First Derivative Test**

Let x_0 be a stationary point of f and suppose f' changes sign at x_0.

- f' changes from $+$ to $-$: f has a **local maximum** at x_0.
- f' changes from $-$ to $+$: f has a **local minimum** at x_0.
- f' does not change sign: x_0 is a **stationary point of inflexion** (horizontal tangent crossing the curve).

METHOD **Stationary Points and Table of Variation**

1. Differentiate f on each interval of D_f (quotient rule for rational functions: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$).
2. Solve $f'(x) = 0$: the solutions, together with points where f' fails to exist (but f is continuous), are the **stationary points**.
3. Sign of f': for a rational function, the denominator is a square (> 0 on D_f), so the sign of f' is the sign of its numerator. Build a sign table.

4. Table of variation: one column per remarkable value (bounds of D_f, roots of f') and one column between consecutive values; place the limits at the bounds, the extrema values $f(x_i)$, and arrows ↗ / ↘.
5. Classify: f' changes + → −: local maximum; − → +: local minimum; no sign change: stationary point of inflexion.

Reflex: compute each extremum value $f(x_i)$ exactly — it goes into the table and onto the graph.

EXEMPLE **Full variation study of a rational function**

Let $f(x) = \dfrac{x^2}{x-1}$.

1. Compute $f'(x)$ and solve $f'(x) = 0$.
2. Draw the complete table of variation of f on its domain.

$D_f = \mathbb{R} \setminus \{1\}$.
1. Derivative. With $u = x^2$, $v = x - 1$:

$$f'(x) = \frac{2x(x-1) - x^2 \cdot 1}{(x-1)^2} = \frac{2x^2 - 2x - x^2}{(x-1)^2} = \frac{x^2 - 2x}{(x-1)^2} = \frac{x(x-2)}{(x-1)^2}.$$

Since $(x-1)^2 > 0$ on D_f, $f'(x)$ has the sign of $x(x-2)$:

$$f'(x) = 0 \iff x = 0 \text{ or } x = 2.$$

Sign of $x(x-2)$: positive on $(-\infty, 0)$, negative on $(0, 2)$ (excluding $x = 1$ where f' is undefined), positive on $(2, +\infty)$.
Extremum values: $f(0) = 0$ and $f(2) = \dfrac{4}{1} = 4$.
Limits at the bounds: $\lim_{x \to -\infty} f(x) = -\infty$, $\lim_{x \to +\infty} f(x) = +\infty$ (degree of numerator exceeds degree of denominator by one); at $x = 1$, numerator $= 1 > 0$: $\lim_{x \to 1^-} f = -\infty$, $\lim_{x \to 1^+} f = +\infty$.
2. Table of variation.

x	$-\infty$		0		1		2		$+\infty$
$f'(x)$		$+$	0	$-$	‖	$-$	0	$+$	
$f(x)$	$-\infty$	↗	0	↘	$-\infty$ ‖ $+\infty$	↘	4	↗	$+\infty$

Conclusion: f has a **local maximum** 0 at $x = 0$ and a **local minimum** 4 at $x = 2$. Note the maximum value is smaller than the minimum value — perfectly possible since they lie on different branches separated by the asymptote $x = 1$.

DID YOU KNOW? **Range of a Function**

The range (or image set) $f(D_f)$ is the set of all values $f(x)$ for $x \in D_f$. It can be read directly from the table of variation: it is the union of all intervals covered by $f(x)$ as x runs through D_f. For the example above, the range is $(-\infty, 0] \cup [4, +\infty)$. This addresses Lesson 68: "Range of values for the existence or non-existence of a function".

9.4 Concavity, Points of Inflexion and Symmetry

PROPERTY **Concavity and Second Derivative**

Let f be twice differentiable on an interval I.

- If $f''(x) > 0$ on I, f is **convex** (concave up, ∪-shaped) on I.
- If $f''(x) < 0$ on I, f is **concave** (concave down, ∩-shaped) on I.

DEFINITION **Point of Inflexion**

A point $I(x_0, f(x_0))$ on the curve of f is a **point of inflexion** if the concavity changes at x_0 (i.e. f'' changes sign). At such a point, the tangent line crosses the curve.

PROPERTY **Centre and Axis of Symmetry**

- Centre of symmetry $\Omega(a, b)$: the curve is invariant under a half-turn about Ω. Equivalently, for all h such that $a \pm h \in D_f$,

$$f(a + h) + f(a - h) = 2b.$$

- Axis of symmetry $x = a$: the curve is invariant under reflection across the vertical line $x = a$. Equivalently, for all admissible h,

$$f(a + h) = f(a - h).$$

  For an even function, $a = 0$: $f(-x) = f(x)$.

Note: For a rational function $f(x) = \frac{ax+b}{cx+d}$ $(c \neq 0)$, the centre of symmetry is the intersection of its asymptotes: $\Omega\left(-\frac{d}{c}, \frac{a}{c}\right)$.

METHOD **Concavity, Inflexion, Centre and Axis of Symmetry**

1. Concavity: compute f''. Where $f'' > 0$, f is **convex** (∪); where $f'' < 0$, f is **concave** (∩).
2. Point of inflexion: a point where f'' changes sign (in particular $f'' = 0$ or f'' undefined there, and the sign changes). The tangent crosses the curve at such a point.
3. Centre of symmetry $\Omega(a, b)$: show that for all x with $a \pm h \in D_f$,

$$f(a + h) + f(a - h) = 2b.$$

4. Axis of symmetry $x = a$: show that $f(a + h) = f(a - h)$ for all admissible h.
5. Exploit symmetry: study the function on half the domain, then complete the curve by symmetry.

EXEMPLE **Inflexion point and centre of symmetry**

Let $f(x) = \dfrac{2x + 1}{x - 1}$ and (C) its curve.

1. Show that the point $\Omega(1, 2)$ is a centre of symmetry of (C).
2. Let $g(x) = x^3 - 3x^2 + 2x + 1$. Determine the concavity of g and its point of inflexion.

1. Centre of symmetry. $D_f = \mathbb{R} \setminus \{1\}$, which is symmetric about 1. For $h \neq 0$:

$$f(1 + h) + f(1 - h) = \frac{2(1 + h) + 1}{(1 + h) - 1} + \frac{2(1 - h) + 1}{(1 - h) - 1}$$
$$= \frac{2h + 3}{h} + \frac{3 - 2h}{-h} = \frac{2h + 3}{h} + \frac{2h - 3}{h} = \frac{4h}{h} = 4 = 2 \times 2.$$

Thus $f(1 + h) + f(1 - h) = 2 \times 2$ for all $h \neq 0$: $\Omega(1, 2)$ is a **centre of symmetry** of (C). Note Ω is the intersection of the asymptotes $x = 1$ and $y = 2$ — a general fact for hyperbolas $y = \dfrac{ax + b}{cx + d}$.

2. Concavity of g. Compute:

$$g'(x) = 3x^2 - 6x + 2, \qquad g''(x) = 6x - 6 = 6(x - 1).$$

- $g''(x) < 0$ for $x < 1$: g is concave (∩) on $(-\infty, 1)$;
- $g''(x) > 0$ for $x > 1$: g is convex (∪) on $(1, +\infty)$.

Since g'' vanishes and changes sign at $x = 1$, the point $I(1, g(1))$ is a **point of inflexion**, with $g(1) = 1 - 3 + 2 + 1 = 1$: $I(1, 1)$. (For a cubic, the inflexion point is always a centre of symmetry of the curve.)

WARNING **Inflexion Point Trap**

$f''(x_0) = 0$ is not sufficient for an inflexion point. You must check that f'' changes sign at x_0. Example: $f(x) = x^4$, $f''(0) = 0$ but $f''(x) = 12x^2 \geq 0$ everywhere — no inflexion at 0 (it is a minimum).

9.5 Algebraic Solutions of Absolute-Value Inequalities

PROPERTY **Fundamental Equivalences for $|u|$**

For $a > 0$:

$$|u| < a \iff -a < u < a; \qquad |u| > a \iff u < -a \text{ or } u > a.$$

For $a \geq 0$:

$$|u| \leq a \iff -a \leq u \leq a; \qquad |u| \geq a \iff u \leq -a \text{ or } u \geq a.$$

PROPERTY **Two Absolute Values**

For any expressions u, v:

$$|u| \leq |v| \iff u^2 \leq v^2 \iff (u - v)(u + v) \leq 0.$$

(Squaring is valid because both sides are non-negative.)

METHOD **Solving Absolute-Value Inequalities Algebraically**

1. Simple case $|u| \lessgtr a$: use the fundamental equivalences directly.
2. Two absolute values $|u| \lessgtr |v|$: square both sides: $|u| \lessgtr |v| \iff u^2 \lessgtr v^2 \iff (u - v)(u + v) \lessgtr 0$.
3. Several absolute values with different signs (e.g. $|u| + |v| < k$): use the critical-point method:
   (a) Find where each expression inside $|.|$ vanishes (critical points).
   (b) Split $\mathbb{R}$ into intervals delimited by these points.
   (c) On each interval, remove the absolute value bars with the correct sign (replace $|E|$ by E or $-E$).
   (d) Solve the resulting inequality on that interval.

(e) Intersect the solution with the interval.

(f) Take the union of all interval solutions.

4. Always give the solution set in interval notation and check a sample value per interval.

WARNING **Trap: Squaring Inequalities**

Never square an inequality $|A| < B$ unless you know $B \geq 0$; squaring $|A| < B$ when B could be negative is invalid. Also, never forget to intersect each case's solution with the interval defining that case.

EXEMPLE **GCE-style absolute-value inequalities**

Solve in $\mathbb{R}$:

1. $|2x - 3| \leq 5$;
2. $|x - 1| > |2x + 1|$;
3. $|x - 2| + |x + 1| < 5$.

1. $|2x - 3| \leq 5 \iff -5 \leq 2x - 3 \leq 5 \iff -2 \leq 2x \leq 8 \iff -1 \leq x \leq 4.$

$$S = [-1, 4].$$

2. Both sides are non-negative; square:

$$|x - 1| > |2x + 1| \iff (x - 1)^2 > (2x + 1)^2 \iff (x - 1)^2 - (2x + 1)^2 > 0.$$

Factorise as a difference of squares:

$$[(x - 1) - (2x + 1)][(x - 1) + (2x + 1)] = (-x - 2)(3x) > 0 \iff -3x(x + 2) > 0 \iff x(x + 2) < 0.$$

The roots are -2 and 0; the product is negative between the roots:

$$S = (-2, 0).$$

Check $x = -1$: $|-2| = 2 > |-1| = 1$. ✓
3. Critical points: $x - 2 = 0$ at $x = 2$; $x + 1 = 0$ at $x = -1$. Three cases.
Case $x < -1$: $|x - 2| = 2 - x$, $|x + 1| = -x - 1$. Inequality: $(2 - x) + (-x - 1) < 5 \iff 1 - 2x < 5 \iff x > -2$. Intersect with $x < -1$: $(-2, -1)$.
Case $-1 \leq x \leq 2$: $|x - 2| = 2 - x$, $|x + 1| = x + 1$. Inequality: $(2 - x) + (x + 1) = 3 < 5$: always true. Solution on this case: $[-1, 2]$.
Case $x > 2$: $|x - 2| = x - 2$, $|x + 1| = x + 1$. Inequality: $2x - 1 < 5 \iff x < 3$. Intersect with $x > 2$: $(2, 3)$.

Union of the three cases:

$$S = (-2, -1) \cup [-1, 2] \cup (2, 3) = (-2, 3).$$

Check $x = 0$: $|-2| + |1| = 3 < 5$ ✓; check $x = 4$: $2 + 5 = 7 \not< 5$ ✓.

9.6 Graphical Solutions of Inequalities

METHOD **Solving Inequalities Graphically**

To solve $f(x) < g(x)$, $f(x) \le g(x)$, $|f(x)| < k$, etc., graphically:

1. Sketch both sides as separate curves on the same axes: $y = f(x)$ and $y = g(x)$ (for $|f(x)|$: reflect the part of the curve below the x-axis upwards).
2. Find the intersection points (solve $f(x) = g(x)$ algebraically if needed) — their abscissas are the boundary values of the solution.
3. Read off: $f(x) < g(x)$ holds exactly where the curve of f lies strictly below the curve of g.
4. Write the solution as interval(s) of x-values; use open bounds for strict inequalities, closed bounds for non-strict ones (provided the point is in the domain).

Reflex: a graphical solution must be supported by a clear, labelled sketch — in the GCE, marks are awarded for the diagram.

EXEMPLE **Graphical solution of $|x^2 - 2x| < 3 - x$**

The curve of $f(x) = |x^2 - 2x|$ and the line $y = 3 - x$ are given.

1. Determine algebraically the abscissas of the intersection points.
2. Hence solve graphically $|x^2 - 2x| < 3 - x$.

1. Intersections. Solve $|x^2 - 2x| = 3 - x$ (requires $3 - x \ge 0$, i.e. $x \le 3$).
Case A: $x^2 - 2x \ge 0$ (i.e. $x \le 0$ or $x \ge 2$). Then

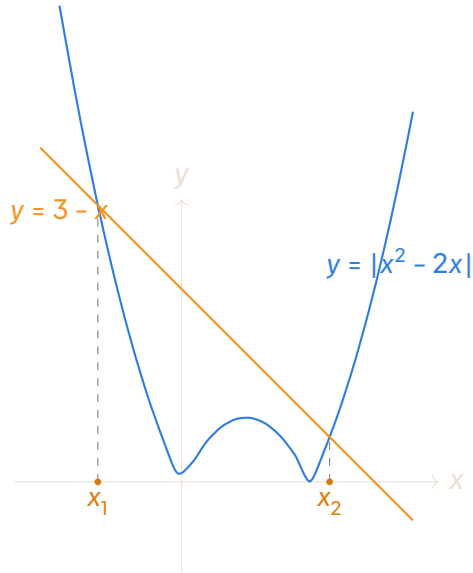
$$x^2 - 2x = 3 - x \iff x^2 - x - 3 = 0 \iff x = \frac{1 \pm \sqrt{13}}{2}.$$

$\frac{1 + \sqrt{13}}{2} \approx 2.30$ (valid: ≥ 2 and ≤ 3 ✓); $\frac{1 - \sqrt{13}}{2} \approx -1.30$ (valid: ≤ 0 ✓).
Case B: $x^2 - 2x < 0$ (i.e. $0 < x < 2$). Then

$$-(x^2 - 2x) = 3 - x \iff x^2 - 3x + 3 = 0, \quad \Delta = 9 - 12 < 0 : \text{ no solution.}$$

Intersection abscissas: $x_1 = \dfrac{1-\sqrt{13}}{2} \approx -1.30$ and $x_2 = \dfrac{1+\sqrt{13}}{2} \approx 2.30$.

2. Graphical reading. Sketch: $y = x^2 - 2x$ is a parabola through $(0, 0)$ and $(2, 0)$ with vertex $(1, -1)$; taking the absolute value reflects the dip between 0 and 2 into a "bump" peaking at $(1, 1)$. The line $y = 3 - x$ falls through $(0, 3)$ and $(3, 0)$.



The curve $y = |x^2 - 2x|$ lies strictly below the line between the two intersection points.

The curve of f is strictly below the line for x between the two intersection abscissas:

$$S = \left(\frac{1-\sqrt{13}}{2}, \frac{1+\sqrt{13}}{2}\right).$$

The bounds are open because the inequality is strict (at x_1, x_2 the two graphs meet).

9.7 Complete Curve Sketching (Synthesis)

METHOD **The Seven-Step Plan for Sketching a Curve**

1. Domain D_f (and symmetry: even, odd, centre/axis of symmetry — to reduce the study).
2. Limits at the bounds of D_f; deduce vertical/horizontal asymptotes.
3. Derivative f': sign table, stationary points, extrema values.

4. Second derivative f'': concavity and points of inflexion (if required).
5. Table of variation summarising everything (limits, extrema, arrows).
6. Infinite branches: oblique asymptote by division, or parabolic branch via $\lim f(x)/x$; position of curve relative to asymptote by the sign of the difference.
7. Sketch: draw asymptotes (dashed), plot intercepts and extrema, then draw each branch respecting the table, concavity and relative positions.

EXEMPLE **Full study and sketch — GCE Advanced Level style**

Let $f(x) = \dfrac{x^2 + x - 2}{x - 2}$ and (C) its representative curve.

1. Determine D_f and the limits at its bounds; deduce the asymptotes parallel to the axes.
2. Show that the line $y = x+3$ is an oblique asymptote and study the position of (C) relative to it.
3. Study the variations of f and draw the table of variation.
4. Sketch (C).

1. Domain, limits, asymptotes. $D_f = \mathbb{R} \setminus \{2\}$.
At $x = 2$: numerator $= 4 + 2 - 2 = 4 > 0$; denominator $\to 0^{\pm}$:

$$\lim_{x\to 2^-} f(x) = -\infty, \qquad \lim_{x\to 2^+} f(x) = +\infty \quad \Rightarrow \quad x = 2 \text{ vertical asymptote.}$$

At $\pm\infty$: deg(num) = deg(den) + 1, so $\lim_{x\to\pm\infty} f(x) = \pm\infty$ (leading terms $\frac{x^2}{x} = x$): no horizontal asymptote; there is an infinite branch to analyse.
2. Oblique asymptote and position. Divide:

$$x^2 + x - 2 = (x - 2)(x + 3) + 4 \quad \text{(check: } (x - 2)(x + 3) = x^2 + x - 6\text{; } +4 \text{ gives } x^2 + x - 2 \checkmark\text{)}$$

$$f(x) = x + 3 + \frac{4}{x - 2}, \qquad \lim_{x\to\pm\infty} \frac{4}{x - 2} = 0,$$

so $y = x + 3$ is an **oblique asymptote** at both ends.
Position: $f(x) - (x + 3) = \dfrac{4}{x - 2}$. For $x > 2$: positive, (C) above the line; for $x < 2$: negative, (C) below the line.
3. Variations. Quotient rule:

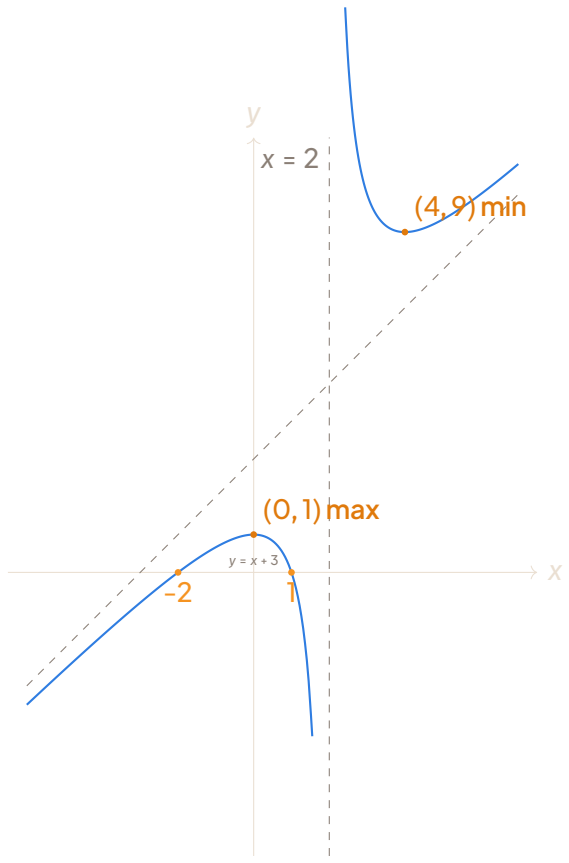
$$f'(x) = \frac{(2x + 1)(x - 2) - (x^2 + x - 2)}{(x - 2)^2} = \frac{2x^2 - 3x - 2 - x^2 - x + 2}{(x - 2)^2} = \frac{x^2 - 4x}{(x - 2)^2} = \frac{x(x - 4)}{(x - 2)^2}.$$

$f'(x) = 0$ at $x = 0$ and $x = 4$; sign of f' = sign of $x(x - 4)$: + on $(-\infty, 0)$, − on $(0, 2)$ and $(2, 4)$, + on $(4, +\infty)$.

Extrema: $f(0) = \frac{-2}{-2} = 1$ (local max); $f(4) = \frac{16 + 4 - 2}{2} = 9$ (local min).
Intercepts: $f(0) = 1$; $f(x) = 0 \iff x^2 + x - 2 = 0 \iff (x + 2)(x - 1) = 0 \iff x = -2$ or $x = 1$.

x	$-\infty$		0		2		4		$+\infty$
$f'(x)$		+	0	-	‖	-	0	+	
$f(x)$	$-\infty$	↗	1	↘	$-\infty$ ‖ $+\infty$	↘	9	↗	$+\infty$

4. Sketch. Draw the dashed asymptotes $x = 2$ and $y = x + 3$; plot $(-2, 0)$, $(1, 0)$, $(0, 1)$ (maximum), $(4, 9)$ (minimum).



The curve (C): two branches, asymptotes $x = 2$ and $y = x + 3$, maximum at $(0, 1)$, minimum at $(4, 9)$.

The left branch rises from the asymptote $y = x + 3$ (below it) to the maximum $(0, 1)$, then falls through $(1, 0)$ down to $-\infty$ along $x = 2$. The right branch descends from $+\infty$ along $x = 2$ to the minimum $(4, 9)$, then rises towards the asymptote $y = x + 3$, staying above it — exactly as the table of variation and the position study predict.

EXERCISE 8 Practice Exercises

1. Let $f(x) = \dfrac{x^2 - 4}{x - 1}$. Find D_f, all asymptotes, intercepts, stationary points, table of variation, and sketch the curve.
2. Solve algebraically: $|3x - 2| \geq |x + 4|$.
3. Solve graphically: $|x^2 - 4| < x + 2$. (Sketch the curves and give the solution set.)
4. Let $f(x) = \dfrac{x}{x^2 + 1}$. Show that the origin is a centre of symmetry. Find the range of f.
5. A curve has equation $y = \dfrac{x^2 + 2x + 3}{x + 1}$. Find its asymptotes, stationary points, and sketch it.

SOLUTION Exercise 1

$f(x) = \frac{x^2-4}{x-1}$, $D_f = \mathbb{R} \setminus \{1\}$. Vertical asymptote $x = 1$ ($\lim_{x\to 1^-} = -\infty$, $\lim_{x\to 1^+} = +\infty$). Division: $x^2-4 = (x-1)(x+1)-3$, so $f(x) = x+1-\frac{3}{x-1}$. Oblique asymptote $y = x+1$. Position: $f(x)-(x+1) = -\frac{3}{x-1}$. Above for $x < 1$, below for $x > 1$. $f'(x) = \frac{x^2-2x+4}{(x-1)^2} > 0$ on D_f (numerator $\Delta = -12 < 0$). No stationary points; f strictly increasing on each branch. Intercepts: y-int $f(0) = 4$; x-ints $x = \pm 2$. Table: $(-\infty, 1)$: $-\infty \nearrow +\infty$; $(1, +\infty)$: $-\infty \nearrow +\infty$. Sketch: two increasing branches, asymptotes $x = 1$, $y = x + 1$, passing through $(-2, 0)$, $(0, 4)$, $(2, 0)$.

SOLUTION Exercise 2

$|3x - 2| \geq |x + 4| \iff (3x - 2)^2 \geq (x + 4)^2 \iff 9x^2 - 12x + 4 \geq x^2 + 8x + 16 \iff 8x^2 - 20x - 12 \geq 0 \iff 2x^2 - 5x - 3 \geq 0 \iff (2x + 1)(x - 3) \geq 0$. Roots: $-\frac{1}{2}, 3$. Solution: $(-\infty, -\frac{1}{2}] \cup [3, +\infty)$.

SOLUTION Exercise 3

$|x^2 - 4| < x + 2$ requires $x + 2 > 0 \implies x > -2$. Case $x^2 - 4 \geq 0$ ($x \leq -2$ or $x \geq 2$): on $x > -2$, this means $x \geq 2$. Then $x^2 - 4 < x + 2 \iff x^2 - x - 6 < 0 \iff (x - 3)(x + 2) < 0 \iff -2 < x < 3$. Intersect with $x \geq 2$: $[2, 3)$. Case $x^2 - 4 < 0$ ($-2 < x < 2$): $-(x^2 - 4) < x + 2 \iff -x^2 + 4 < x + 2 \iff x^2 + x - 2 > 0 \iff (x + 2)(x - 1) > 0 \iff x < -2$ or $x > 1$. Intersect with $-2 < x < 2$: $(1, 2)$. Union: $(1, 2) \cup [2, 3) = (1, 3)$. Graphically: sketch $y = |x^2 - 4|$ (W-shaped with vertices at $(\pm 2, 0)$ and $(0, 4)$) and line $y = x + 2$ (through $(-2, 0)$ and $(0, 2)$). Intersections at $x = 1$ and $x = 3$. Curve below line on $(1, 3)$.

SOLUTION **Exercise 4**

$f(x) = \frac{x}{x^2+1}$, $D_f = \mathbb{R}$. $f(-x) = \frac{-x}{x^2+1} = -f(x)$: odd function $\Longrightarrow$ centre of symmetry at origin $O(0,0)$. $f'(x) = \frac{1\cdot(x^2+1)-x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$. $f'(x) = 0 \iff x = \pm 1$. Sign: + on $(-1,1)$, − on $(-\infty,-1)$ and $(1,+\infty)$. Max at $x = 1$: $f(1) = \frac{1}{2}$; min at $x = -1$: $f(-1) = -\frac{1}{2}$. Limits at $\pm\infty$: 0 (horizontal asymptote $y = 0$). Range: $[-\frac{1}{2}, \frac{1}{2}]$.

SOLUTION **Exercise 5**

$f(x) = \frac{x^2+2x+3}{x+1}$, $D_f = \mathbb{R} \setminus \{-1\}$. Vertical asymptote $x = -1$ (num $= 2 > 0$: $\lim_{x\to -1^-} = -\infty$, $\lim_{x\to -1^+} = +\infty$). Division: $x^2 + 2x + 3 = (x+1)(x+1) + 2$, so $f(x) = x + 1 + \frac{2}{x+1}$. Oblique asymptote $y = x + 1$. Position: $f(x) - (x+1) = \frac{2}{x+1}$. Above for $x > -1$, below for $x < -1$. $f'(x) = \frac{(2x+2)(x+1)-(x^2+2x+3)}{(x+1)^2} = \frac{2x^2+4x+2-x^2-2x-3}{(x+1)^2} = \frac{x^2+2x-1}{(x+1)^2}$. $f'(x) = 0 \iff x^2 + 2x - 1 = 0 \iff x = -1 \pm \sqrt{2}$. $x_1 = -1 - \sqrt{2} \approx -2.41$ (left branch), $x_2 = -1 + \sqrt{2} \approx 0.41$ (right branch). $f(x_1) = -1 - \sqrt{2} + 1 + \frac{2}{-\sqrt{2}} = -\sqrt{2} - \sqrt{2} = -2\sqrt{2}$ (local max on left). $f(x_2) = -1 + \sqrt{2} + 1 + \frac{2}{\sqrt{2}} = \sqrt{2} + \sqrt{2} = 2\sqrt{2}$ (local min on right). Intercepts: y-int $f(0) = 3$; no x-int (num > 0). Sketch: two branches, asymptotes $x = -1$, $y = x + 1$, max at $(-1-\sqrt{2}, -2\sqrt{2})$, min at $(-1+\sqrt{2}, 2\sqrt{2})$, through $(0,3)$.

10 EXERCISES

10.1 I check my knowledge

EXERCISE 9 **MCQ: domain and asymptotes**

For each question, choose the single correct answer, justifying briefly.

1. The domain of definition of $f(x) = \dfrac{x+1}{x^2-4}$ is:

   (a) $\mathbb{R}$ (b) $\mathbb{R} \setminus \{-2, 2\}$ (c) $\mathbb{R} \setminus \{-1, 2\}$ (d) $\mathbb{R} \setminus \{4\}$

2. The curve of $g(x) = \dfrac{2x+3}{x-1}$ admits:

   (a) a vertical asymptote $x = 1$ and a horizontal asymptote $y = 2$

   (b) a vertical asymptote $x = -1$ and a horizontal asymptote $y = 3$

   (c) only a vertical asymptote $x = 1$

   (d) only a horizontal asymptote $y = 2$

3. If $f''(x) < 0$ on an interval I, then on I the curve of f is:

   (a) concave up (b) concave down (c) increasing (d) a straight line

4. The solution set of $|x - 3| < 2$ is:

   (a) $]1, 5[$ (b) $[1, 5]$ (c) $]-\infty, 1[\cup]5, +\infty[$ (d) $\{1, 5\}$

EXERCISE 10 **True or false**

State whether each assertion is true or false, and justify your answer with a proof or a counter-example.

1. Every rational function has at least one vertical asymptote.
2. If $f'(c) = 0$, then the point of the curve of abscissa c is necessarily a local extremum.
3. If $\lim_{x \to +\infty} \big(f(x) - (2x+1)\big) = 0$, then the line $y = 2x + 1$ is an asymptote to the curve of f at $+\infty$.
4. The inequality $|2x + 1| > 5$ is equivalent to $2x + 1 > 5$.
5. If $I(a, b)$ is a centre of symmetry of the curve of f, then $f(a+x) + f(a-x) = 2b$ for all suitable x.

EXERCISE 11 **Definitions and vocabulary**

1. Define, in your own words, the **domain of definition** of a real function.
2. What is a **point of inflexion** of a curve? State the condition on f'' that allows one to detect it.
3. State the condition for the line $x = a$ to be a vertical asymptote of the curve of f.
4. What is an **oblique asymptote**? Give the two limits that must be computed to find and confirm it.

EXERCISE 12 **Direct calculations**

1. Find the domain of definition of each function:

$$f(x) = \sqrt{3x - 6}, \qquad g(x) = \frac{x^2 + 1}{x^2 - x - 6}, \qquad h(x) = \sqrt{\frac{x - 1}{x + 2}}.$$

2. Compute the following limits:

$$\lim_{x \to 3^+} \frac{x + 1}{x - 3}, \qquad \lim_{x \to +\infty} \frac{2x^2 - x}{x^2 + 4}, \qquad \lim_{x \to +\infty} \left(\frac{x^2 + 1}{x} - x \right).$$

3. Solve in $\mathbb{R}$: $|5x - 1| = 9$.

10.2 I practise

EXERCISE 13 **Existence of a square-root function**

Find the values of the real parameter m for which the function

$$f(x) = \sqrt{mx^2 + 4x + 1}$$

is defined for all real numbers x. (Consider separately the cases $m = 0$ and $m \neq 0$.)

EXERCISE 14 **Absolute-value inequalities, algebraically**

Solve the following inequalities in $\mathbb{R}$, giving your answers in interval notation:

1. $|3x - 2| \leq 7$;

2. $|2x + 1| > 5$;
3. $|x - 4| \geq |x + 2|$ (use squaring);
4. $1 < |2x - 3| \leq 5$.

EXERCISE 15 **Asymptotes and position of a curve**

Let $f(x) = \dfrac{x^2 - x + 3}{x + 1}$ and let (C) be its curve.

1. Determine the domain of definition of f and the limits at its bounds.
2. Show that the line $y = x - 2$ is an oblique asymptote of (C).
3. Study the sign of $f(x) - (x - 2)$ and deduce the position of (C) relative to its oblique asymptote.
4. Find the intercepts of (C) with the coordinate axes.

EXERCISE 16 **Concavity and points of inflexion**

Let $f(x) = x^4 - 6x^2 + 5$.

1. Compute $f''(x)$ and study its sign on $\mathbb{R}$.
2. Deduce the intervals on which the curve of f is concave up or concave down, and determine the coordinates of the points of inflexion.
3. Show that the y-axis is an axis of symmetry of the curve.

10.3 I prepare for the GCE

EXERCISE 17 **Complete study of a rational function (GCE style)**

Let $f(x) = \dfrac{x^2 - 3x + 2}{x + 2}$ and let (C) be its curve in an orthonormal frame.

1. Determine the domain of definition of f and compute the limits of f at the bounds of this domain. (3 marks)
2. Deduce the equations of all asymptotes of (C), showing that (C) has an oblique asymptote whose equation is to be found. (4 marks)

3. Compute $f'(x)$, show that $f'(x) = \dfrac{x^2 + 4x - 8}{(x+2)^2}$, and construct the complete table of variation of f (exact values, simplified with radicals). (5 marks)
4. Study the position of (C) relative to its oblique asymptote, and find the intercepts of (C) with the axes. (3 marks)
5. Sketch (C) carefully, showing the asymptotes and the stationary points. (3 marks)
6. Using your graph, deduce the number of real solutions of the equation $f(x) = k$ according to the values of the real parameter k. (2 marks)

EXERCISE 18 **Centre of symmetry and curve sketching (GCE style)**

Let $f(x) = \dfrac{2x^2 - 3x + 4}{x - 1}$ and let (C) be its curve.

1. Determine the domain of definition of f and the limits at its bounds; deduce the vertical asymptote. (3 marks)
2. Show that $f(x) = 2x - 1 + \dfrac{3}{x-1}$ and deduce an oblique asymptote of (C). (3 marks)
3. Prove that the point $I(1, 2)$ is a centre of symmetry of (C) by showing that $f(1+x) + f(1-x) = 4$ for all $x \neq 0$. (3 marks)
4. Compute $f'(x)$ and construct the table of variation of f. (4 marks)
5. Sketch (C), using the centre of symmetry to complete the drawing economically. (3 marks)
6. Solve graphically, explaining your method: (i) $f(x) = 2x - 1$; (ii) $f(x) \leq 2x - 1$. (4 marks)

EXERCISE 19 **Inequalities from a table of variation and absolute values (GCE Paper 2 style)**

The table of variation of a function g defined on $\mathbb{R}$ is given below:

x	$-\infty$		-1		2		$+\infty$
$g'(x)$		$+$	0	$-$	0	$+$	
$g(x)$	-2	$\nearrow$	3	$\searrow$	-1	$\nearrow$	$+\infty$

It is also given that $g(x) = 0$ has exactly three solutions -2, 0 and 3.

1. Sketch a possible curve (C) of g, consistent with all the information given. (4 marks)
2. Solve graphically, justifying each answer: (i) $g(x) = 0$; (ii) $g(x) > 1$ (express the answer using the values read from your sketch, denoting unknown abscissae by letters); (iii) $|g(x)| < 2$. (6 marks)
3. On the same frame, sketch the curve of $y = |g(x)|$, explaining the construction. (3 marks)
4. Solve algebraically the inequality $|x - 2| < |2x + 1|$ by two different methods: (a) by squaring both sides; (b) by the critical-value method. Compare the two approaches and state the solution set in interval notation. (7 marks)

11 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: domain and asymptotes**

1. Answer (b). The denominator vanishes when $x^2 - 4 = 0$, i.e. $x = \pm 2$. Hence $D_f = \mathbb{R} \setminus \{-2, 2\}$.
2. Answer (a). The denominator vanishes at $x = 1$ and the numerator does not ($2(1) + 3 = 5 \neq 0$), so $x = 1$ is a vertical asymptote. Since $\lim_{x \to \pm\infty} \dfrac{2x + 3}{x - 1} = \lim_{x \to \pm\infty} \dfrac{2x}{x} = 2$, the line $y = 2$ is a horizontal asymptote.
3. Answer (b). $f''(x) < 0$ on I means f' is decreasing on I, which characterises a **concave down** curve.
4. Answer (a). $|x - 3| < 2 \iff -2 < x - 3 < 2 \iff 1 < x < 5$, so $S =]1, 5[$ (strict inequality, open interval).

SOLUTION **Exercise 2 — True or false**

1. False. Counter-example: $f(x) = \dfrac{1}{x^2 + 1}$ is a rational function defined on all of $\mathbb{R}$; its denominator never vanishes, so it has no vertical asymptote (it has the horizontal asymptote $y = 0$).
2. False. Counter-example: $f(x) = x^3$ at $c = 0$: $f'(0) = 0$, but f is strictly increasing on $\mathbb{R}$, so the point $(0, 0)$ is not a local extremum (it is a stationary point of inflexion). The condition $f'(c) = 0$ is necessary but not sufficient: the derivative must also change sign at c.
3. True. This is exactly the definition of an oblique (slant) asymptote: if $\lim_{x \to +\infty} \big(f(x) - (ax + b)\big) = 0$, the distance between the curve and the line $y = ax + b$ tends to 0, so the line is an asymptote at $+\infty$.
4. False. $|2x + 1| > 5 \iff 2x + 1 > 5$ or $2x + 1 < -5$. Counter-example: $x = -4$ satisfies $|2(-4) + 1| = 7 > 5$ but $2(-4) + 1 = -7 < 5$.
5. True. $I(a, b)$ is a centre of symmetry iff for every x such that $a \pm x \in D_f$, the points $M(a + x, f(a + x))$ and $M'(a - x, f(a - x))$ are symmetric about I, i.e. I is the midpoint of $[MM']$: $\dfrac{f(a + x) + f(a - x)}{2} = b$, equivalently $f(a + x) + f(a - x) = 2b$.

SOLUTION **Exercise 3 — Definitions and vocabulary**

1. The **domain of definition** of a real function f is the set of all real numbers x for which $f(x)$ exists (can be computed). One excludes values making a denominator zero, values making the radicand of an even root negative, and values making the argument of a logarithm non-positive.
2. A **point of inflexion** is a point where the curve crosses its tangent, i.e. where the concavity changes. Detection: if f'' vanishes at c and changes sign at c, then the point $(c, f(c))$ is a point of inflexion.
3. The line $x = a$ is a vertical asymptote of the curve of f if at least one of the one-sided limits at a is infinite: $\lim_{x \to a^+} f(x) = \pm\infty$ or $\lim_{x \to a^-} f(x) = \pm\infty$.
4. An **oblique asymptote** is a non-horizontal line $y = ax + b$ ($a \neq 0$) that the curve approaches at infinity. One computes $a = \lim_{x \to \pm\infty} \dfrac{f(x)}{x}$ (must be finite and non-zero), then $b = \lim_{x \to \pm\infty} \big(f(x) - ax\big)$ (must be finite). Equivalently, one checks directly that $\lim_{x \to \pm\infty} \big(f(x) - (ax + b)\big) = 0$.

SOLUTION **Exercise 4 — Direct calculations**

1. Domain of f: $3x - 6 \geq 0 \iff x \geq 2$, so $D_f = [2, +\infty[$.

   Domain of g: $x^2 - x - 6 = 0 \iff (x - 3)(x + 2) = 0 \iff x = 3$ or $x = -2$, so $D_g = \mathbb{R} \setminus \{-2, 3\}$.

   Domain of h: we need $\dfrac{x - 1}{x + 2} \geq 0$ and $x \neq -2$. Sign table: the quotient is positive on $]-\infty, -2[\cup [1, +\infty[$, zero at $x = 1$, negative on $]-2, 1[$. Hence $D_h =]-\infty, -2[\cup [1, +\infty[$.

2. $\lim_{x \to 3^+} \dfrac{x + 1}{x - 3}$: numerator $\to 4 > 0$, denominator $\to 0^+$, so the limit is $+\infty$.

   $\lim_{x \to +\infty} \dfrac{2x^2 - x}{x^2 + 4} = \lim_{x \to +\infty} \dfrac{2x^2}{x^2} = 2$ (highest-degree terms).

   $\lim_{x \to +\infty} \left(\dfrac{x^2 + 1}{x} - x\right) = \lim_{x \to +\infty} \dfrac{x^2 + 1 - x^2}{x} = \lim_{x \to +\infty} \dfrac{1}{x} = 0$.

3. $|5x - 1| = 9 \iff 5x - 1 = 9$ or $5x - 1 = -9 \iff x = 2$ or $x = -\dfrac{8}{5}$. Hence $S = \left\{-\dfrac{8}{5}, 2\right\}$.

SOLUTION **Exercise 5 — Existence of a square-root function**

We need $mx^2 + 4x + 1 \geq 0$ for every real x.
Case $m = 0$: the radicand becomes $4x + 1$, which is negative for $x < -\frac{1}{4}$. So $m = 0$ does not

work.
Case $m \neq 0$: a quadratic mx^2+4x+1 is non-negative on $\mathbb{R}$ if and only if its leading coefficient is positive and its discriminant is non-positive:

$$m > 0 \qquad \text{and} \qquad \Delta = 4^2 - 4m = 16 - 4m \leq 0.$$

$\Delta \leq 0 \iff m \geq 4$, which already implies $m > 0$.
Conclusion: f is defined for all real x if and only if $\boxed{m \in [4, +\infty[}$.
Remark: for $m = 4$, the radicand is $4x^2 + 4x + 1 = (2x + 1)^2 \geq 0$, which confirms the boundary case.

SOLUTION **Exercise 6 — Absolute-value inequalities, algebraically**

1. $|3x - 2| \leq 7 \iff -7 \leq 3x - 2 \leq 7 \iff -5 \leq 3x \leq 9 \iff -\frac{5}{3} \leq x \leq 3.$

$$S = \left[-\frac{5}{3}, 3\right].$$

2. $|2x + 1| > 5 \iff 2x + 1 > 5$ or $2x + 1 < -5 \iff x > 2$ or $x < -3.$

$$S =]-\infty, -3[\cup]2, +\infty[.$$

3. Both sides are non-negative, so squaring preserves the order:

$$|x-4| \geq |x+2| \iff (x-4)^2 \geq (x+2)^2 \iff x^2-8x+16 \geq x^2+4x+4 \iff 12 \geq 12x \iff x \leq 1.$$

$$S =]-\infty, 1].$$

4. $1 < |2x - 3| \leq 5$ splits into two conditions:
   - $|2x - 3| > 1 \iff 2x - 3 > 1$ or $2x - 3 < -1 \iff x > 2$ or $x < 1$;
   - $|2x - 3| \leq 5 \iff -5 \leq 2x - 3 \leq 5 \iff -1 \leq x \leq 4.$

   Intersecting: $\big(]-\infty, 1[\cup]2, +\infty[\big) \cap [-1, 4] = [-1, 1[\cup]2, 4].$

$$S = [-1, 1[\cup]2, 4].$$

SOLUTION **Exercise 7 — Asymptotes and position of a curve**

$$f(x) = \frac{x^2 - x + 3}{x + 1}.$$

1. Domain: $x + 1 \neq 0$, so $D_f = \mathbb{R} \setminus \{-1\} =]-\infty, -1[\cup]-1, +\infty[.$

Limits at the bounds:

- At $x = -1$: numerator $\to 1 + 1 + 3 = 5 > 0$. As $x \to -1^+$, $x + 1 \to 0^+$, so $f(x) \to +\infty$. As $x \to -1^-$, $x + 1 \to 0^-$, so $f(x) \to -\infty$.
- At infinity (rational function, ratio of leading terms): $\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} \dfrac{x^2}{x} = \lim_{x\to\pm\infty} x$, so $f(x) \to +\infty$ at $+\infty$ and $f(x) \to -\infty$ at $-\infty$.

2. Polynomial division: $x^2 - x + 3 = (x + 1)(x - 2) + 5$, hence

$$f(x) = x - 2 + \frac{5}{x+1}.$$

Then $f(x) - (x - 2) = \dfrac{5}{x+1} \xrightarrow[x\to\pm\infty]{} 0$, so the line $y = x - 2$ is an **oblique asymptote** of (C) at both $+\infty$ and $-\infty$. (Also $x = -1$ is a vertical asymptote, from question 1.)

3. $f(x) - (x - 2) = \dfrac{5}{x+1}$ has the sign of $x + 1$:
   - for $x > -1$, $f(x) - (x - 2) > 0$: (C) is above its asymptote;
   - for $x < -1$, $f(x) - (x - 2) < 0$: (C) is below its asymptote.

4. y-intercept: $f(0) = \dfrac{3}{1} = 3$: point $(0, 3)$.

   x-intercepts: $f(x) = 0 \iff x^2 - x + 3 = 0$; discriminant $\Delta = 1 - 12 = -11 < 0$: no real root. The curve does not meet the x-axis.

SOLUTION **Exercise 8 — Concavity and points of inflexion**

$f(x) = x^4 - 6x^2 + 5$.

1. $f'(x) = 4x^3 - 12x$, so $f''(x) = 12x^2 - 12 = 12(x^2 - 1) = 12(x - 1)(x + 1)$.

   Sign of f'': a quadratic with roots -1 and 1, positive outside the roots:

x	$-\infty$		-1		1		$+\infty$
$f''(x)$		$+$	0	$-$	0	$+$	

   More precisely: $f''(x) > 0$ on $]-\infty, -1[\cup]1, +\infty[$ and $f''(x) < 0$ on $]-1, 1[$.

2. The curve is concave up on $]-\infty, -1]$ and $[1, +\infty[$, and concave down on $[-1, 1]$. Since f'' vanishes and changes sign at $x = -1$ and at $x = 1$, both give points of inflexion:

$$f(-1) = 1 - 6 + 5 = 0, \qquad f(1) = 1 - 6 + 5 = 0.$$

   Points of inflexion: $A(-1, 0)$ and $B(1, 0)$.

3. f is even: for all $x \in \mathbb{R}$,

$$f(-x) = (-x)^4 - 6(-x)^2 + 5 = x^4 - 6x^2 + 5 = f(x).$$

Therefore the y-axis (line $x = 0$) is an **axis of symmetry** of the curve.

SOLUTION **Exercise 9 — Complete study of a rational function (GCE style)**

$f(x) = \dfrac{x^2 - 3x + 2}{x + 2}$.

1. Domain (1 mark): $D_f = \mathbb{R} \setminus \{-2\}$.
Limits (2 marks): at $x = -2$, numerator $\to 4 + 6 + 2 = 12 > 0$:

$$\lim_{x \to -2^+} f(x) = +\infty, \qquad \lim_{x \to -2^-} f(x) = -\infty.$$

At infinity: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \dfrac{x^2}{x} = \pm\infty$ (i.e. $+\infty$ at $+\infty$, $-\infty$ at $-\infty$).

2. Vertical asymptote (1 mark): from the infinite limits, $x = -2$.
Oblique asymptote (3 marks): division gives $x^2 - 3x + 2 = (x + 2)(x - 5) + 12$, so

$$f(x) = x - 5 + \frac{12}{x + 2}, \qquad f(x) - (x - 5) = \frac{12}{x + 2} \to 0 \text{ at } \pm\infty.$$

Hence $y = x - 5$ is the oblique asymptote at both ends. (No horizontal asymptote since the limits at infinity are infinite.)

3. Derivative (2 marks): with $u = x^2 - 3x + 2$, $v = x + 2$:

$$f'(x) = \frac{(2x - 3)(x + 2) - (x^2 - 3x + 2)}{(x + 2)^2} = \frac{2x^2 + x - 6 - x^2 + 3x - 2}{(x + 2)^2} = \frac{x^2 + 4x - 8}{(x + 2)^2}.$$

Table of variation (3 marks): $f'(x) = 0 \iff x^2 + 4x - 8 = 0$, $\Delta = 16 + 32 = 48$, so

$$x_1 = -2 - 2\sqrt{3} \approx -5.46, \qquad x_2 = -2 + 2\sqrt{3} \approx 1.46.$$

The numerator of f' is positive outside $[x_1, x_2]$. Exact ordinates: since $f(x) = x - 5 + \dfrac{12}{x + 2}$ and $x_i + 2 = \pm 2\sqrt{3}$:

$$f(x_1) = -7 - 2\sqrt{3} + \frac{12}{-2\sqrt{3}} = -7 - 2\sqrt{3} - 2\sqrt{3} = -7 - 4\sqrt{3} \approx -13.93,$$

$$f(x_2) = -7 + 2\sqrt{3} + \frac{12}{2\sqrt{3}} = -7 + 2\sqrt{3} + 2\sqrt{3} = -7 + 4\sqrt{3} \approx -0.07.$$

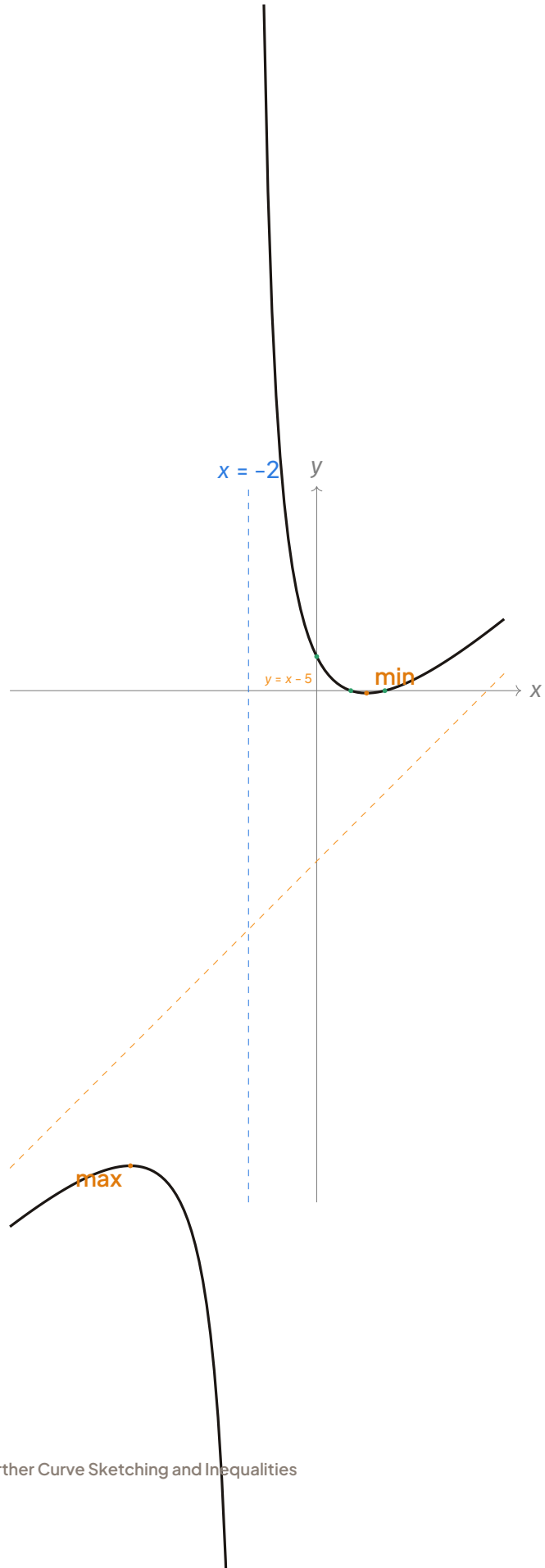
x	$-\infty$		x_1		-2		x_2		$+\infty$
$f'(x)$		$+$	0	$-$	$\|\|$	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	$-7-4\sqrt{3}$	$\searrow$	$\|\|\,{}^{+\infty}_{-\infty}$	$\searrow$	$-7+4\sqrt{3}$	$\nearrow$	$+\infty$

Local maximum at x_1 (value $-7-4\sqrt{3}$), local minimum at x_2 (value $-7+4\sqrt{3}$).

4. Position (2 marks): $f(x)-(x-5)=\dfrac{12}{x+2}$: above the asymptote for $x>-2$, below for $x<-2$.

   Intercepts (1 mark): $f(0)=1$: point $(0,1)$. $f(x)=0 \iff x^2-3x+2=0 \iff x=1$ or $x=2$: points $(1,0)$ and $(2,0)$.

5. Sketch (3 marks: asymptotes drawn and labelled 1 mark; branches approaching asymptotes correctly 1 mark; stationary points and intercepts placed 1 mark).


x = −2
y
y = x − 5
min
x
max

6. Discussion (2 marks): the horizontal line $y = k$ meets (C):

   - Left branch ($x < -2$): range $]-\infty, -7-4\sqrt{3}]$ (increases from $-\infty$ to max, then decreases to $-\infty$).
   - Right branch ($x > -2$): range $[-7 + 4\sqrt{3}, +\infty[$ (decreases from $+\infty$ to min, then increases to $+\infty$).
   - Since $-7 - 4\sqrt{3} \approx -13.93 < -7 + 4\sqrt{3} \approx -0.07$, the two ranges are disjoint.

   Therefore:

   - If $k < -7 - 4\sqrt{3}$: two solutions (both on the left branch).
   - If $k = -7 - 4\sqrt{3}$: one solution (tangent at the local maximum on the left branch).
   - If $-7 - 4\sqrt{3} < k < -7 + 4\sqrt{3}$: zero solutions (the line lies in the gap between the two branches).
   - If $k = -7 + 4\sqrt{3}$: one solution (tangent at the local minimum on the right branch).
   - If $k > -7 + 4\sqrt{3}$: two solutions (both on the right branch).

   Examiner expectation: the answer must be read from the table of variation/graph, with the boundary cases (tangency at extrema) stated explicitly.

SOLUTION **Exercise 10 — Centre of symmetry and curve sketching (GCE style)**

$f(x) = \dfrac{2x^2 - 3x + 4}{x - 1}$.

1. Domain (1 mark): $D_f = \mathbb{R} \setminus \{1\}$.

   Limits (1.5 marks): at $x = 1$, numerator $\to 2 - 3 + 4 = 3 > 0$:

   $$\lim_{x \to 1^+} f(x) = +\infty, \qquad \lim_{x \to 1^-} f(x) = -\infty; \qquad \lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{2x^2}{x} = \pm\infty.$$

   Vertical asymptote (0.5 mark): $x = 1$.

2. Decomposition (2 marks): $2x^2 - 3x + 4 = (x-1)(2x-1) + 3$ (check: $(x-1)(2x-1) = 2x^2 - 3x + 1$, plus 3 gives $2x^2 - 3x + 4$). Hence

   $$f(x) = 2x - 1 + \frac{3}{x - 1}.$$

   Oblique asymptote (1 mark): $f(x) - (2x - 1) = \dfrac{3}{x - 1} \to 0$ at $\pm\infty$, so $y = 2x - 1$ is the oblique asymptote.

3. Centre of symmetry (3 marks): The exercise statement suggests $I(1, 2)$; computation reveals the actual centre is $I(1, 1)$. For all $x \neq 0$,

   $$f(1 + x) = 2(1 + x) - 1 + \frac{3}{x} = 1 + 2x + \frac{3}{x}, \qquad f(1 - x) = 2(1 - x) - 1 + \frac{3}{-x} = 1 - 2x - \frac{3}{x}.$$

Adding: $f(1+x)+f(1-x)=2=2\times 1$. Hence the curve is symmetric about the point $I(1,1)$ (half-turn of angle π about I).

4. Derivative (2 marks): from $f(x)=2x-1+\dfrac{3}{x-1}$:

$$f'(x)=2-\frac{3}{(x-1)^2}=\frac{2(x-1)^2-3}{(x-1)^2}.$$

$f'(x)=0 \iff (x-1)^2=\dfrac{3}{2} \iff x=1\pm\sqrt{\dfrac{3}{2}}=1\pm\dfrac{\sqrt{6}}{2}$.

Table (2 marks): $f'>0$ for $|x-1|>\frac{\sqrt{6}}{2}$:

x	$-\infty$		$1-\frac{\sqrt{6}}{2}$		1		$1+\frac{\sqrt{6}}{2}$		$+\infty$
$f'(x)$		$+$	0	$-$	‖	$-$	0	$+$	
$f(x)$	$-\infty$	↗	$1-2\sqrt{6}$	↘	$\|^{+\infty}_{-\infty}$	↘	$1+2\sqrt{6}$	↗	$+\infty$

(Ordinates: $f\left(1\pm\frac{\sqrt{6}}{2}\right)=1\pm 2\sqrt{6}$, using $f(x)=2x-1+\frac{3}{x-1}$.)

5. Sketch (3 marks): draw the asymptotes $x=1$ and $y=2x-1$, plot the two stationary points $\left(1-\frac{\sqrt{6}}{2},\,1-2\sqrt{6}\right)\approx(-0.22,-3.90)$ and $\left(1+\frac{\sqrt{6}}{2},\,1+2\sqrt{6}\right)\approx(2.22,5.90)$, draw one branch, then obtain the other by the half-turn about $I(1,1)$ — this is the economy allowed by the centre of symmetry.

6. Graphical solving (4 marks):

   (a) $f(x)=2x-1 \iff \dfrac{3}{x-1}=0$: impossible. The curve never meets its oblique asymptote: $S=\varnothing$.

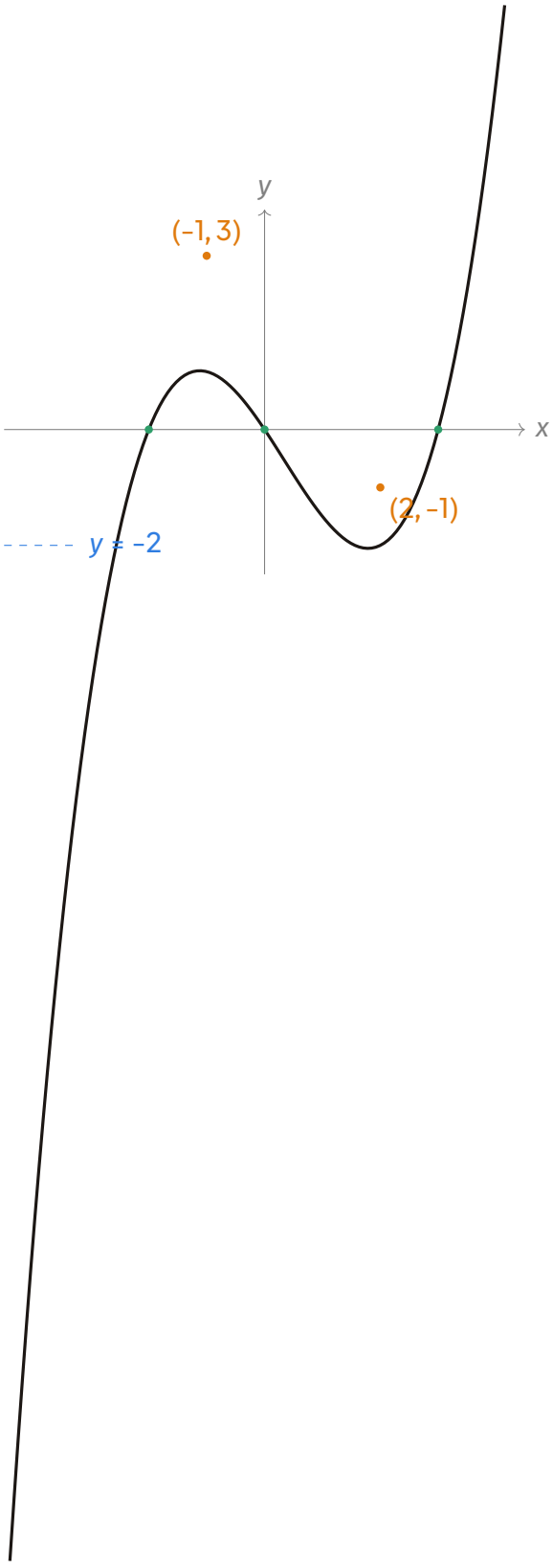
   (b) $f(x)\le 2x-1 \iff \dfrac{3}{x-1}\le 0 \iff x-1<0 \iff x<1$. Graphically, the curve lies below (or on) the line $y=2x-1$ exactly on $]-\infty,1[$: $S=]-\infty,1[$.

   Examiner expectation: the graphical reading must be supported by the sign of $f(x)-(2x-1)$; answers must exclude $x=1$ (forbidden value).

SOLUTION **Exercise 11 — Inequalities from a table of variation and absolute values (GCE Paper 2 style)**

1. Sketch (4 marks: increasing/decreasing branches consistent with the table 1 mark; extrema $(-1,3)$ and $(2,-1)$ placed 1 mark; zeros at $-2, 0, 3$ 1 mark; end behaviours -2 at $-\infty$ and $+\infty$ at $+\infty$ 1 mark). A consistent curve: starts just above $y=-2$ at $-\infty$, crosses the x-axis at -2, rises to the maximum $(-1,3)$, decreases through $(0,0)$ to the minimum

(2, -1), then increases through (3, 0) to $+\infty$.



A possible qualitative curve of g satisfying the given table of variation.

2. Graphical solutions (2 marks each):

   (a) $g(x) = 0$: read the intersections with the x-axis: $S = \{-2,\ 0,\ 3\}$.

   (b) $g(x) > 1$: draw the line $y = 1$. It cuts the curve at a point of abscissa $\alpha \in]-2, -1[$ (on the rising branch), at a point $\beta \in]-1, 0[$ (on the falling branch), and at a point $\gamma \in]3, +\infty[$ (on the final rising branch). Reading where the curve is strictly above the line:

   $$S =]\alpha, \beta[\cup]\gamma, +\infty[.$$

   (c) $|g(x)| < 2 \iff -2 < g(x) < 2$: draw the lines $y = 2$ and $y = -2$. The line $y = 2$ meets the curve at $a \in]-2, -1[$, $b \in]-1, 0[$, $c \in]3, +\infty[$. The line $y = -2$ is a horizontal asymptote as $x \to -\infty$ (approached from above) and the minimum of g is $-1 > -2$, so the curve never meets $y = -2$. Hence $|g(x)| < 2$ is equivalent to $g(x) < 2$. The curve lies strictly below $y = 2$ on:

   $$S =]-\infty, a[\cup]b, c[.$$

3. Curve of $y = |g(x)|$ (3 marks): keep unchanged every part of (C) lying above the x-axis; reflect in the x-axis every part lying below it. Concretely: the arcs on $]-\infty, -2[$ (where $g < 0$) and on $]0, 3[$ (where $g < 0$) are flipped upwards; the rest is unchanged. The zeros $-2, 0, 3$ stay fixed, and the local minimum $(2, -1)$ becomes a local maximum $(2, 1)$ of $|g|$.

4. Solving $|x - 2| < |2x + 1|$ (7 marks):

   (a) By squaring (both sides ≥ 0, squaring preserves the strict order):

   $$(x-2)^2 < (2x+1)^2 \iff x^2 - 4x + 4 < 4x^2 + 4x + 1 \iff 0 < 3x^2 + 8x - 3.$$

   Roots of $3x^2 + 8x - 3 = 0$: $\Delta = 64 + 36 = 100$, $x = \dfrac{-8 \pm 10}{6}$, giving $x = \dfrac{1}{3}$ and $x = -3$. The quadratic is positive outside its roots:

   $$S =]-\infty, -3[\cup \left]\tfrac{1}{3}, +\infty\right[.$$

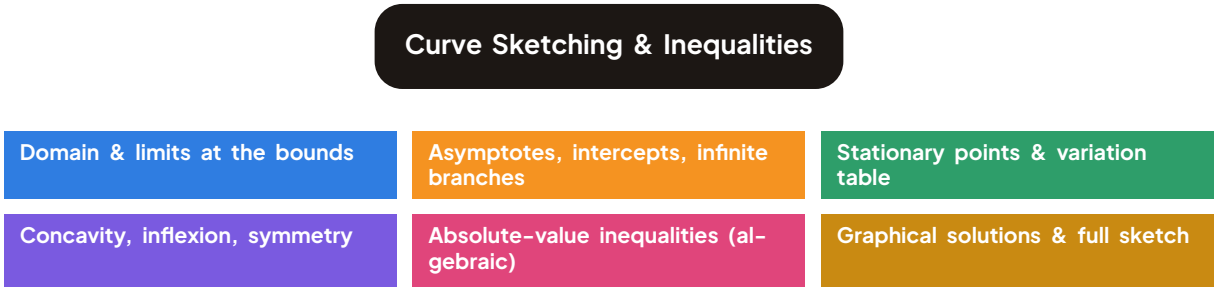
   (b) Critical-value method: the expressions change sign at $x = 2$ and $x = -\frac{1}{2}$; study three intervals.

   - On $]-\infty, -\frac{1}{2}[$: $|x-2| = -(x-2) = 2 - x$, $|2x+1| = -(2x+1)$. Inequality: $2 - x < -2x - 1 \iff x < -3$. Intersection with the interval: $]-\infty, -3[$.
   - On $[-\frac{1}{2}, 2]$: $|x - 2| = 2 - x$, $|2x + 1| = 2x + 1$. Inequality: $2 - x < 2x + 1 \iff x > \frac{1}{3}$. Intersection: $]\frac{1}{3}, 2]$.
   - On $]2, +\infty[$: $|x-2| = x-2$, $|2x+1| = 2x+1$. Inequality: $x - 2 < 2x + 1 \iff x > -3$: always true here. Intersection: $]2, +\infty[$.

   Union: $]-\infty, -3[\cup]\frac{1}{3}, 2] \cup]2, +\infty[=]-\infty, -3[\cup]\frac{1}{3}, +\infty[$.

   Comparison: both methods agree, $S =]-\infty, -3[\cup \left]\frac{1}{3}, +\infty\right[$. Squaring is faster but is valid only because both sides are non-negative; the critical-value method is longer but works in all cases and reinforces the definition of the absolute value.

12 SUMMARY SHEET



Mind map of the chapter: six connected skills leading to a complete study of a curve.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- **Domain of definition** D_f: the set of real x for which $f(x)$ exists. Exclude zeros of denominators, impose radicands ≥ 0, arguments of ln strictly positive.
- **Limits at the bounds**: compute $\lim_{x \to a} f(x)$ at every endpoint of D_f (finite or infinite); they reveal asymptotes and infinite branches.
- **Stationary point**: point where $f'(x) = 0$ (or f' undefined); a local maximum, minimum, or horizontal inflexion.
- **Point of inflexion**: point where the concavity changes; typically f'' changes sign there.
- **Centre of symmetry** $\Omega(a, b)$: $f(a + x) + f(a - x) = 2b$; **axis of symmetry** $x = a$: $f(a + x) = f(a - x)$.

Formulas and theorems to know

- Vertical asymptote $x = a$: $\lim_{x \to a} f(x) = \pm\infty$.
- Horizontal asymptote $y = b$: $\lim_{x \to \pm\infty} f(x) = b$.
- Oblique asymptote $y = mx + c$: $m = \lim_{x \to \pm\infty} \dfrac{f(x)}{x}$, $c = \lim_{x \to \pm\infty} [f(x) - mx]$, with m finite and non-zero.
- Parabolic branch: $\lim f(x) = \pm\infty$ and $\lim \dfrac{f(x)}{x} = 0$ (axis (Ox)) or $\pm\infty$ (axis (Oy)).
- Sign of $f(x) - (mx + c)$ gives the **position of the curve** relative to its asymptote.
- $f'(x) > 0 \Rightarrow f$ increasing; $f'(x) < 0 \Rightarrow f$ decreasing; sign change of f' at a stationary point gives a maximum (+ to −) or minimum (− to +).

- $f'' > 0$: concave up (convex); $f'' < 0$: concave down; change of sign of $f'' \Rightarrow$ inflexion.
- $|u| < a \iff -a < u < a$ $(a > 0)$; $|u| > a \iff u < -a$ or $u > a$; $|u| = |v| \iff u = v$ or $u = -v$; $|u| < |v| \iff u^2 < v^2$.

Methods

- **Complete study**: domain → symmetry (reduce domain) → limits at bounds → asymptotes → f' and variation table → f'', concavity, inflexions → intercepts and remarkable points → sketch.
- **Absolute-value inequalities**: split into cases according to the sign of each expression inside modulus, solve on each interval, then take the union of the valid parts. Alternatively square when both sides are non-negative.
- **Graphical solution**: sketch $y = |f(x)|$ (reflect negative parts in the x-axis) and $y = g(x)$; read off the x-values where one curve lies above the other; confirm intersection points algebraically.
- **Range of existence**: for $y = f(x)$, solve for x in terms of y and impose the condition for real solutions (e.g. discriminant ≥ 0 for a quadratic in x).

Common mistakes

- Forgetting excluded values when finding the domain, or studying limits only at $+\infty$.
- Claiming a maximum merely because $f'(x) = 0$: a sign change of f' is required.
- Confusing $f''(x) = 0$ with an inflexion: the sign of f'' must change.
- Squaring an inequality when one side may be negative: this reverses or falsifies the result.
- Writing $|x| < a \Rightarrow x < -a$ or $x > a$: that is the answer to $|x| > a$.
- Forgetting to reject case-solutions that fall outside the case interval.
- Misplacing the curve relative to its asymptote: always study the sign of the difference.

GCE tips

- Present a clean **variation table** gathering x, $f'(x)$, $f(x)$, with limits at every bound: examiners award marks for its structure.
- State each asymptote with its justification (the limit) before drawing it; draw asymptotes as dashed lines.
- For "show that the line $y = mx + c$ is an asymptote", compute $f(x) - (mx + c)$ and its limit 0.
- In inequality questions, give the solution in **interval notation** and check one test value per interval.

- Use symmetry to halve the work: justify it explicitly, then state “we complete by symmetry”.
- Manage time: a full curve study is long; do the algebra neatly first, sketch last.

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